

The Yang–Mills workbench

a reader of its results, with proofs

SU(2) lattice Yang–Mills: vacuum, energy, spectral bounds and continuum investigations
(Zenodo record 10.5281/zenodo.22883643)

An audit and reading prepared by Claude (Anthropic),
model `claude-opus-5-5` (Opus 5.5), at maximum reasoning effort

Workbench: KokunoYumeto, with the AI contributors and the sources credited below

27 September 2026 (version 2, after one independent referee pass)

Companion to the reader of the split-zero zeta programme, version 2, DOI 10.5281/zenodo.22977243

Abstract

The Zenodo record 10.5281/zenodo.22883643, *SU(2) lattice Yang–Mills: vacuum, energy, spectral bounds and continuum investigations*, is the Yang–Mills part of the owner’s research workbench (KokunoYumeto/yang-mills-interacting-workbench). It holds 43 files from five editions, dated 8 to 21 September 2026. The record’s front file, `README.md`, is still the one written for the first edition of 8 September.

This reader says:

- which of the 43 files is current and which are superseded;
- what each document claims, and on which regime;
- what is proved, with proofs;
- how far each statement was checked.

The central result (Section 3) concerns the Kogut–Susskind Hamiltonian of $SU(2)$ lattice gauge theory on open cubic boxes. It is an explicit lower bound on the spectral gap above the ground state, on the whole gauge-invariant space, uniform in the box size and the lattice spacing, for bare coupling $g^2 \geq 3.6836$ in the workbench’s normalisation. The threshold rests on computed coefficient bounds of orders 3 to 5. With the order-2 inputs recomputed in the first-pass audit, the same argument gives a gap of at least 1.52κ for $g^2 \geq 3.9781$, which includes the workbench’s benchmark $g^2 = 4$.

Three consequences of the record’s material are added here:

- the gap persists at complex coupling, so the ground state depends analytically on the coupling on a disk whose radius does not depend on the box;
- an upper bound on the gap: for $g^2 \geq 13$, for example, the gap lies between 2.90κ and 3.0003κ ;
- a gap bound on the full space of states, not only the gauge-invariant one.

Section 4 states the heat-flow results of the latest edition:

- every constant of the box-independent heat remainder is re-derived here, and its prefactor $67896/169$ is improved to $2829/13$ by a sharper estimate of the mean;
- the benchmark table for relaxation into the whole complementary physical space is recomputed exactly from the stated inputs, and then sharpened.

Sections 5–6 map the earlier editions and the Jacobian-map, Navier–Stokes and S^6 material, and Section 7 states the obstructions with their exact scope.

All results hold at fixed lattice spacing and strong bare coupling. The workbench does not claim a continuum mass gap, and nothing here bears on that problem.

About this reader

The workbench, and who made it.

- The record’s README.md (“Source history, corrections and attribution”) says that the human participant proposed the research directions and possible connections, and that Codex developed, checked and corrected the written constructions. It attributes the public stewardship to KokunoYumeto, the owner of the workbench.
- The repository’s README describes the workbench as open and AI-assisted, including work with ChatGPT 5.6 Sol and GPT-6 Astra.
- The workbench credits:
 - the exponential form of the vacuum, and its character and Casimir framework, to Schütte, Zheng and Hamer [10];
 - the Fourier-algebra estimates to Eymard [3];
 - semigroup generation to Lumer and Phillips [7];
 - the creation method behind the quantum line’s first volume-uniform gap to Yarotsky [12], with Baez’s spin-network framework [2] as its reference for the physical space (quantum line, p. 19; the record’s README.md lists both).
- Two inputs of side branches are credited to Levent Alpöge by the workbench’s attribution file (ATTRIBUTEION.md; record file ALPOGE_ATTRIBUTEION_20260921.md). One is a polynomial map that the file identifies with the counterexample to the Jacobian conjecture announced by Alpöge on 20 July 2026, whose announcement, as the file records, credits Akhil for the question and Fable for the work. The other is the construction of a complex threefold, circulated by Alpöge and, as the file says, produced with Claude [1]. The file says that neither is used as a hypothesis of the 21 September heat and volume proofs. That the fifth-reference results (F1–F42, H1–H34) do not use them either follows from reading those proofs (Sections 3 and 4).
- The Navier–Stokes material rests on OpenAI’s manuscript *Finite Time Blowup for Navier–Stokes* [8], which the workbench cites and does not claim to have verified.

What was read.

- The repository at commit fa79faf (21 September 2026), the head of origin/main on 26 September.
- The record’s 43 files. Twenty-three of them are byte-identical to files of that commit (checked by MD5). The other twenty were downloaded from the record on 26 September, except the 87 MB Overleaf archive (file 08_), and their MD5 checksums equal the record’s.
- The two proof manuscripts of the latest edition in full: FIFTH_REFERENCE.md (F1–F42) and HEAT_AND_COMPLEMENT.md (H1–H34).
- For the other documents, their scope and conclusion sections, and the passages cited below.

The audit notes 21_, 28_, 34_, 46_, 46a_ and 47_, and the reader of four workbenches [11], are on the branch `claude/claude-ab-grind-20260925` of KokunoYumeto/zeta-function-research-reader, in `contrib/claude-ab/grind_20260925/`.

How it was checked.

- Section 3 comes from the Yang–Mills section of the reader of four workbenches, which was refereed three times, after a referee pass on its notes (Appendix B).
- The new material of Section 4 was checked by this reader’s author, with a new script. It recomputes the constants from the manuscript’s formulas and the stated inputs. The constants 8, 32, $\frac{1}{8}$, 30 and 48 are derived in the text.

- This version includes the findings of one independent referee pass, by a Claude instance that wrote none of the text. It read the reader for overstatement, for understatement and missed generality, and for missed or implied results. Each finding was verified before it was applied, by reading the record at the cited place or by an independent script (Appendix B).
- Everything else carries a status label.

Status labels.

- Proved here: the proof is given in full in this reader.
- Generalised here: the record’s statement is proved here under weaker hypotheses, or with a sharper constant, than the record states.
- Audited: the workbench’s proof was read in full, every step was checked, and independent checks were run.
- Audited in part: the parts named were checked; the rest is located.
- Reproduced: the finite content of a statement (its constants, tables or counts) was recomputed by an independent script from the stated inputs; the general argument was not re-derived.
- Conditional: proved, given inputs that were not checked here; the inputs are named.
- Located: the statement and its proof were found and are described; nothing was checked.

The workbench itself separates written analytic arguments from finite exact-arithmetic certificates and from numerical diagnostics, and says so at each result; its own qualifications are quoted where they matter.

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1 The problem, the setting, and what the workbench claims

1.1 The continuum problem

The Clay Mathematics Institute’s problem, described by Jaffe and Witten [4], asks for a proof that for every compact simple gauge group a nontrivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. The existence part asks for a continuum theory satisfying axioms at least as strong as the Wightman or Osterwalder–Schrader axioms ([4], §4); estimates for a model at fixed lattice spacing do not by themselves answer it.

1.2 The lattice setting of the workbench

The workbench works with a spatial lattice and continuous time, in the Hamiltonian formulation of Kogut and Susskind [6], for the group $SU(2)$.

- *Box*. The vertices form the open box $\{-L, \dots, L\}^3$, $L \geq 2$. The links e are the positively oriented nearest-neighbour edges inside the box, and the plaquettes p are the elementary faces inside it; there are $M_L = 12L^2(2L + 1)$ of them.
- *Space*. Each link carries a matrix $U_e \in SU(2)$, and the configuration space carries the product Haar probability measure. The *physical* Hilbert space is the subspace of L^2 invariant under the gauge transformations at every vertex, boundary vertices included.
- *Hamiltonian* (the workbench’s normalisation):

$$H_L = \kappa K + \kappa \tilde{\zeta} (2M_L - S), \quad K = - \sum_{e,a} X_{e,a}^2, \quad S = \sum_p W_p, \quad \kappa = \frac{2g^2}{a}, \quad \tilde{\zeta} = \frac{1}{4g^4}.$$

Here $X_{e,a}$ differentiates U_e along e^{iT_a} with $T_a = -i\sigma_a/2$, so K is the sum of the link Casimirs; W_p is the trace of the ordered product of the four links around p ; $a > 0$ is the lattice spacing and $g > 0$ the bare coupling. The plaquette term equals $(2g^2a)^{-1} \sum_p (2 - W_p) \geq 0$.

- *Gap*. Δ_L is the distance from the ground energy E_0 to the rest of the spectrum of H_L on the physical space. At $\tilde{\zeta} = 0$ it equals 3κ : one plaquette in the fundamental representation, four links of Casimir $\frac{3}{4}$.

Large g is strong coupling, where $\tilde{\zeta}$ is small; the continuum limit needs $a \rightarrow 0$ with $g \rightarrow 0$, the opposite regime. The workbench records the conversion to the convention of Schütte, Zheng and Hamer [10]: $g_C = 2^{3/4}g$, and the Hamiltonians differ by the factor $\sqrt{2}$ and a constant. Numerical thresholds below are in the workbench’s normalisation.

1.3 What is known

- *Strong coupling*. Mass gaps at strong coupling are classical for lattice gauge theories in the Euclidean setting (Osterwalder and Seiler [9]; cited, not re-read here).
- *The exponential vacuum*. Writing the ground state as e^v and solving for v order by order in the coupling is the coupled-cluster method of Hamiltonian lattice field theory; the workbench credits it, with the character and Casimir calculus it uses, to Schütte, Zheng and Hamer [10].

Whether an explicit bound of the kind proved in Section 3 (uniform in the box, on the whole physical space, with a closed coupling threshold) is in the literature was not searched.

1.4 What the workbench claims, and what it does not

The workbench’s front pages are explicit (repository README.md; the readers of 21 September):

- its results are finite-regulator statements, at fixed lattice spacing and on stated coupling domains;

- it does not claim a solution of the four-dimensional continuum mass-gap problem, nor independent certification of the S^6 or Navier–Stokes constructions it uses;
- it separates written analytic arguments from finite exact-arithmetic certificates and from numerical diagnostics, and claims no Lean or independent external analytical certification.

The latest edition also records that its standing continuum path, $a_n = a_0 2^{-n}$ with $g_n^2 = 1/c_n$ and $c_n = g_0^{-2} + \beta n \log 2$, eventually leaves the domain of its estimates when $\beta > 0$ (Proposition 3.9). In the parts read here these scope statements are accurate. Where a statement of the workbench was found to be stronger or weaker than its proof, this reader says so.

1.5 The approach in one paragraph

The ground state of H_L is positive, so it can be written $\psi = e^v / \|e^v\|$. Inserting this into the eigenvalue equation gives a nonlinear equation for v , namely $v = \xi v_1 + B(v, v)$ with $v_1 = S/3$, whose formal solution is a power series $\sum_n \xi^n v_n$ (Lemma 3.4). The workbench computes the first five coefficients exactly, bounds them in weighted Fourier-algebra norms that are local, so that nothing grows with the box, and solves for the remainder by a contraction argument. Returning to the eigenvalue equation then gives a lower bound for every excitation energy (Proposition 3.5). The same control of the vacuum gives box-independent bounds for heat correlations of plaquettes, and bounds for how much the plaquette family mixes with the rest of the physical space (Section 4).

2 The record, and what each document contains

2.1 Five editions under one concept DOI

The record 10.5281/zenodo.22883643 is the latest of five versions of the concept record 10.5281/zenodo.22666540. Each version kept the files of the one before and added new ones, so the latest version holds all 43 files. The version history, read from the record’s public metadata on 26 September 2026:

Version (DOI 10.5281/zenodo....)	Files added	What was added
22666541, 8 Sept. ("2026.09.08")	6	The first checkpoint: a 46-page quantum manuscript (01_), a 72-page spatial manuscript (02_), their sources (03_), a sealed author packet (04_), the record’s README.md and a manifest
22667605, 8 Sept. ("full-workbench")	7	Later snapshots of the quantum and spatial lines (05_, 53 pp.; 06_, 108 pp.) and the first volume manuscript (07_, 62 pp.); the whole Overleaf workbench (08_, 87 MB); a reading map, an edition guide and a manifest (09_–11_)
22678364, 9 Sept. ("consolidated")	10	The five foundation readers (§2.2); the sources of 9 September; an index for machine reading; manifests; a reading guide
22803564, 21 Sept. ("heat-volume")	14	The 297-page reader of the 14–16 September continuation; the 49-page reader of 17 September; the 46-page heat and volume reader of 21 September with its sources; two cumulative archives; reading guides; the attribution clarification
22883643, 21 Sept. ("fifth-reference")	6	The 24-page fifth-reference reader, its LaTeX, Markdown and sources, a cumulative archive, and its reading guide

Two facts matter for a visitor.

- The record’s front file README.md was written for the first version, and it describes the 8 September checkpoint. The current guide is YM_READING_GUIDE_20260921_FIFTH.md, and the current reader is yang_mills_fifth_reference_20260921.pdf.

- Twenty-three of the files are byte-identical to files in the repository at commit fa79faf (checked by MD5, Appendix B). The other twenty are not files of the repository, but five of them occur, byte-identical, inside archives committed there: `AI_READING_INDEX.md`, `READER_PROVENANCE.json` and `SOURCE_MANIFEST.json` in `yang_mills_current_sources_2026-09-09.zip`, and `ALPOGE_ATTRIBUTION_20260921.md` and `YM_READING_GUIDE_20260921.md` in `yang_mills_heat_volume_reader_sources_20260921.zip`.

2.2 The documents, line by line

The mathematical documents form three lines of manuscripts from 8–9 September, each with several dated snapshots, followed by four readers of later continuations. In each line the section list of the latest snapshot contains those of the earlier ones, and the 9 September guide says that the quantum reader is cumulative.

Document	Files in the record	Content, in the document’s own terms
<i>Quantum line.</i> “Extensive true-vacuum quantum blocking in the full spatial Wilson box” (8–9 Sept.)	01_ (46 pp.), 05_ (53 pp.), <code>quantum_nonabelian_vertex</code> (63 pp.), <code>quantum_interacting_tensor_band</code> (71 pp.), <code>quantum_coarse_graining</code> (79 pp., cumulative)	Exact blocking maps of the finite interacting Hamiltonian; a uniform single-link estimate for the actual vacuum; zero-shift reconstruction and comparison of low-energy states; the scope of the spectral conclusion (§11); the material-tensor transfer of the Jacobian map into local-energy states (§§13–16, with Theorems 13.1 and 14.2); a nonzero non-Abelian vertex (§17); the Navier–Stokes curvature-state map (§§20–22)
<i>Spatial line.</i> “The retained cusp, spatial refinement, and the spectral measure of the actual magnetic vacuum translation” (8–9 Sept.)	02_ (72 pp.), 06_ (108 pp.), <code>spatial_continuum</code> (129 pp.)	Joint geometric and coupling sequences built from the S^6 period data; spectral escape and its failure modes; weak-coupling variational bounds; exact plaquette-to-loop maps and a pure-gauge configuration limit; covariance and magnetic-mass lower bounds; a fluid-to-curvature map; §27, the collapse of the finite-box weak-coupling disk (Proposition 7.1); §28, the current conclusion
<i>Volume line.</i> “Volume-independent local estimates in the full $SU(2)$ Wilson vacuum” (8–9 Sept.)	07_ (62 pp.), <code>volume_uniform_vacuum</code> (81 pp.)	Local projections and comparison operators; local electric moments at every positive coupling; uniform local Wilson and electric covariances; cluster coordinates; the gauge-restricted free threshold 3κ ; negative-period collapse; the product remainder on the cusp at $0 < \xi \leq 10^{-16}$; the logarithmic coupling path
Web continuation, 14–16 Sept.	<code>yang_mills_web_continuation</code> (297 pp.) and its sources	The source equations, gauge-coordinate maps, cubic calculation and spectral return; the first gap theorems (order 1 and order 2; Section 5)
Quartic cube reader, 17 Sept.	<code>yang_mills_quartic_cube_reader</code> (49 pp.) and its sources	The fourth-order vacuum source in two presentations, equal as polynomials; the sixth-order energy with the cube contribution $-83/1944$; a gap bound for $g^2 \geq 15/4$
Heat and volume reader, 21 Sept.	<code>yang_mills_heat_volume_20260921</code> (46 pp.) and archives	Fourth-order heat correlations; a box-independent heat remainder on the circle $ \xi < 3/256$; a fixed-spacing spatial limit; the plaquette family’s coupling to the whole complementary space
Fifth-reference reader , 21 Sept.	<code>yang_mills_fifth_reference_20260921</code> (24 pp.) and archive	The latest results: the gap bound of Section 3 and the heat and complement bounds of Section 4, with both proof manuscripts (F1–F42, H1–H34)

Bold marks the current document of each line. The cumulative archives of 21 September also hold the continuation of 18–19 September (sixth source, eighth energy, a 240-face response; directory 20260918-sixth-source); the workbench says that its new spatial tables and final execution package were not delivered (repository README.md).

2.3 How the documents describe their own scope

- The 9 September readers are finite-regulator calculations. The spatial reader’s conclusion (§28) says that the actual interacting continuum construction, and nonzero low-energy spectral weight outside its proved exclusions, remain unresolved, and that no missing covariance, tightness or convergence statement was assumed.
- The quantum reader’s §11 records that rewriting a coarse kinetic coefficient at a “running coupling” is only a coefficient dictionary, not a renormalisation of the original Hamiltonian.
- The volume reader’s cusp estimates hold at fixed coupling $0 < \xi \leq 10^{-16}$, that is $g^2 \geq 5 \cdot 10^7$; its §24 treats the logarithmic coupling path without a small- ξ gap.
- Every reader of 17 and 21 September states that no four-dimensional continuum mass gap is claimed.

These statements were checked against the documents where this reader cites them. The results of the 8–9 September lines were not otherwise audited here; Section 5 lists the few that were.

3 The strong-coupling gap

This section states the central result of the record, from the fifth-reference manuscript FIFTH_REFERENCE.md (equations F1–F42; reproduced as Appendix A of the fifth-reference reader), with its proof. It was first stated in the reader of four workbenches [11], where it had three referee passes (Appendix B).

3.1 The statement

Theorem 3.1 (the fifth-reference gap bound; F26–F42). *Let (m_i, t_i) , $i = 1, \dots, 5$, be the ten rational numbers of the workbench’s F26, $\ell_5(x) = \sum (m_i + 4t_i)x^i$, $\delta_5(x) = \sum_{i,j \leq 5, i+j \geq 6} 3(m_i t_j + m_j t_i)x^{i+j}$, $\mathcal{D}_5 = (1 - \ell_5)^2 - \frac{8}{3}\delta_5$, α_5 the first positive root of $\mathcal{D}_5$, and*

$$d_5(x) = \frac{3}{2}(1 + \sqrt{\mathcal{D}_5(x)}) + \sum_{i=1}^5 (\frac{3}{2}m_i - 6t_i)x^i.$$

If the (m_i, t_i) bound the local norms of the vacuum coefficients as the workbench’s computations assert, then for every $L \geq 2$, every $a > 0$ and every $0 < \xi \leq \alpha_5$,

$$\Delta_L \geq \kappa d_5(\xi).$$

In the coupling,

$$\xi \leq \alpha_5 \iff g^2 \geq \frac{1}{2\sqrt{\alpha_5}} = 3.6835519839857273 \dots, \quad \alpha_5 = 0.0184249535761176166818 \dots;$$

d_5 decreases on $[0, \alpha_5]$ from $d_5(0) = 3$ (at $\xi = 0$, $\Delta_L = 3\kappa$) to $d_5(\alpha_5) = 1.5453 \dots$; so $\Delta_L \geq 1.5453 \kappa$ for all $g^2 \geq 3.6836$, and, as $d_5(1/64) = 1.9068301567 \dots$, $\Delta_L > 1.9068 \kappa$ for all $g^2 \geq 4$, uniformly in the box and the spacing. Equivalently, $q_{H_L - E_0}(\psi f) \geq \kappa d_5(\xi) \text{Var}_{\psi^2}(f)$ for f in the physical form domain, ψ the normalised ground state (the workbench’s form, F37).

Status. Conditional on the computed coefficient bounds of orders 3, 4 and 5 for the threshold 3.6836 and the constants d_5 (the audit’s inventory label is YM-01); its qualitative content, a gap on the whole physical space, uniform in the box and the spacing, above an explicit coupling, holds with audited inputs for $g^2 \geq 3.9781$ (Corollary 3.6(b)).

- Order 1 is verified by hand.
- Order 2 was recomputed independently in the first pass from explicit $SU(2)$ coefficient tensors. The exact values $m(v_2) = 134.0673 \dots$ and $t(v_2) = 19.7953 \dots$ lie below the workbench's 5834/39 and 137/6.
- Order 5 (662 representatives, 124,864 anchored multisets, 5,726 rational primal/dual certificates) was reproduced by re-running the workbench's producer and auditor, with byte-identical output. The workbench's execution record reports these runs; its publication intake matched their hashes but did not repeat them (INTAKE_REVIEW.md). The logic of order 5 was read, not re-implemented.
- Orders 3–4 are inherited from earlier continuations and pinned by hash.

The analytic argument is audited. α_5 , the threshold and d_5 were recomputed by exact bisection and by root isolation, and the monotonicity of d_5 on a 401-point grid.

3.2 The proof: two lemmas and the analytic core

Lemma 3.2 (gauge-invariance Casimir inequality). *Let the links form a finite graph without loops (parallel links allowed) in which every cycle has length at least g ; the cubic box has $g = 4$. On every Peter–Weyl block j that contains a nonzero gauge-invariant vector, $\sum_f j_f \geq g j_e$ for every link e . Hence $c_j \geq \frac{3g}{2} j_e$ for every e , $\sum_e j_e \leq \frac{2}{3} c_j$, and $c_j \geq \frac{3g}{4}$ on the nonconstant physical space; for the box, $c_j \geq 6j_e$ and $c_j \geq 3$. The first inequality holds for all real weights $j_f \geq 0$ that satisfy the vertex inequalities $j_f \leq \sum_{f' \ni v, f' \neq f} j_{f'}$, and it is an equality for $j = \frac{1}{2}$ on a cycle of length g through e and 0 elsewhere.*

Status. Generalised here: the workbench's F7 (audited) is the case $g = 4$ on the cubic lattice, proved there from the absence of triangles; the sharp constant on any graph is its girth. Checked exhaustively over half-integer spins with vertex singlets on the cube graph (1,012 nonzero assignments with every $j_e \leq 1$; constant 4), $K_{3,3}$ (4), the 5-cycle and the Petersen graph (5), the 6-cycle (6) and a triangle (3); and by linear programming over real weights with the vertex inequalities only, on 13 graphs of girth 3 to 8 (among them the Petersen, Heawood, McGee and Tutte–Coxeter graphs, the dodecahedron and the box $\{-1, 0, 1\}^3$), on three parallel links (2), and on the bridge between two triangles. On every link that lies on a shortest cycle the optimum is the girth; on the bridge, which lies on no cycle, it is 4, larger than the girth 3. So the girth is the best constant uniform over the links; Remark 3.3 gives the exact constant for each link.

Proof. A block contains a nonzero invariant exactly when, at every vertex v , the tensor product of the representations of spins j_f ($f \ni v$) contains the trivial representation; by the Clebsch–Gordan rule this gives the vertex inequalities, which are all that is used. *A matching at each vertex.* At v , the inequalities say that no j_f exceeds half of $J_v = \sum_{f \ni v} j_f$. That is the condition for a transport plan between the links at v , with supplies and demands j_f and the diagonal forbidden, by Gale's supply–demand theorem: a set of two or more columns can be served by every row, and a single column f by the rows $f' \neq f$, which carry $J_v - j_f \geq j_f$. Symmetrising the plan gives a symmetric $M^v \geq 0$ with zero diagonal and row sums j_f . *A circulation.* Give each orientation (dart) of a link f the weight $j_f/2$, and let the dart arriving at v along f pass to the dart leaving v along $f' \neq f$ with weight $M_{ff'}^v/2$. The row and column sums of M^v make this a circulation on the darts, which is therefore a nonnegative combination $\sum_C w_C [C]$ of cyclic dart sequences C , that is, closed walks in which consecutive links (cyclically) differ. Counting weights, $\sum_C w_C |C| = \sum_f j_f$ and $\sum_C w_C n_e(C) = j_e$, where $n_e(C)$ is the number of traversals of e by C . *Each walk.* Cut a walk C with $n_e(C) \geq 1$ at its traversals of $e = uv$. Each piece avoids e and is nonempty, since two traversals of e are never consecutive and $u \neq w$. A piece that ends at the endpoint of e where it started is a closed walk in which consecutive links differ, so it contains a cycle and has length at least g ; a piece joining u and w contains a path, which with e forms a cycle, so it has length at least $g - 1$. Hence $|C| \geq g n_e(C)$, and summing, $\sum_f j_f \geq g j_e$. *The Casimir.* Every nonzero spin is at least $\frac{1}{2}$, so $j_f(j_f + 1) \geq \frac{3}{2} j_f$; hence $c_j \geq \frac{3}{2} \sum_f j_f$, which is $\sum_f j_f \leq \frac{2}{3} c_j$, and $c_j \geq \frac{3g}{2} j_e$. On a nonconstant physical block some $j_e \geq \frac{1}{2}$, so

$c_j \geq \frac{3g}{4}$. For the equality case, $j = \frac{1}{2}$ on a cycle of length g satisfies every vertex inequality and has $\sum_f j_f = \frac{g}{2} = g j_e$. $\square$

Remark 3.3 (the exact constant for each link). For a link $e = uw$ let g_e be the length of a shortest cycle through e ($g_e = \infty$ if there is none), and ℓ_x the length of a shortest closed walk at x in the graph without e in which consecutive links differ, except possibly the last and the first. Then $\sum_f j_f \geq \gamma_e j_e$ with $\gamma_e = \min(g_e, 1 + \frac{1}{2}(\ell_u + \ell_w)) \geq g$, and γ_e is attained by half-integer spins with vertex singlets.

Status. Proved here; found by the referee pass of this reader. The referee’s linear programs give γ_e on every link of ten graphs, including both bridge examples.

Proof. Cut each walk C of the circulation at its traversals of e , as in the proof of Lemma 3.2. A piece joining u and w has length at least $g_e - 1$. A piece that returns to its starting endpoint x is a closed walk at x in the graph without e , of length at least ℓ_x . After a piece that returns to w the next traversal of e runs from w to u , and conversely; so, as C is closed, the returning pieces come in pairs, one at u and one at w . Hence $|C| \geq n_e(C)\gamma_e$, and summing, $\sum_f j_f \geq \gamma_e j_e$. For attainment, spin $\frac{1}{2}$ on a shortest cycle through e gives the ratio g_e . For the other value, take $C = eP_w e^{-1}P_u$, with P_w, P_u shortest walks of the kind defining ℓ_w, ℓ_u , and put $j_f = \frac{1}{2}n_f(C)$. Consecutive links of C differ, which gives the vertex inequalities. The spins at a vertex sum to its number of visits by C , an integer, so a singlet exists. The ratio is $|C|/n_e(C) = 1 + \frac{1}{2}(\ell_u + \ell_w)$. $\square$

Lemma 3.4 (ground-state transform). Let $\Gamma(f, h) = \sum_{e,a} (X_{e,a} f)(X_{e,a} h)$, P_H the Haar mean and $Q_H = I - P_H$. For real or complex v , e^v is an eigenfunction of H_L with eigenvalue E iff $Kv = \zeta S + Q_H \Gamma(v, v)$ and $E = \kappa(2M_L \zeta - P_H \Gamma(v, v))$; then $(H_L - E)(e^v h) = \kappa e^v (Kh - 2\Gamma(v, h))$ for every h .

Status. Proved here; the workbench’s F33–F36, audited. The identities are algebraic, so they hold for complex v and complex ζ ; only the positivity of e^v , used for the ground state, needs v real.

Proof. From $X^2(e^v h) = e^v(h(Xv)^2 + 2(Xv)(Xh) + hX^2v + X^2h)$ one gets $K(e^v h) = e^v(h(Kv - \Gamma(v, v)) - 2\Gamma(v, h) + Kh)$. With $h = 1$, $H_L e^v = \kappa e^v(Kv - \Gamma(v, v) + \zeta(2M_L - S))$, which equals $E e^v$ exactly when $Kv - \Gamma(v, v) - \zeta S$ is the constant $E/\kappa - 2M_L \zeta$; taking Haar means ($P_H Kv = 0$, $P_H S = 0$) identifies the constant as $-P_H \Gamma(v, v)$. Inserting this into the formula for $K(e^v h)$ gives the last identity. $\square$

The ground state is unique and positive, so a solution v of the vacuum equation gives the ground state $e^v / \|e^v\|$. With $v_1 = S/3$ (each W_p has Casimir 3) and $B = K^{-1}Q_H \Gamma$ the equation is $v = \zeta v_1 + B(v, v)$, whose formal solution is $\sum \zeta^n v_n$ with $v_n = \sum_{i+j=n} B(v_i, v_j)$; this exponential (coupled-cluster) ansatz is the one the workbench credits, for Hamiltonian lattice field theory, to Schütte, Zheng and Hamer [10].

The analytic core (the workbench’s F4–F32; audited). In the Fourier-algebra norm $\|f\|_X = \sum_j \|A_j(f)\|_1$, which is submultiplicative (Eymard [3]), and in local weighted norms $m(\cdot)$, $t(\cdot)$, $\|\cdot\|_{\text{loc}}$ anchored at a link, $\|B(f, h)\|_{\text{loc}} \leq 3(m(f)t(h) + m(h)t(f)) \leq \frac{2}{3}\|f\|_{\text{loc}}\|h\|_{\text{loc}}$, by Lemma 3.2. With $q_5 = \sum_{i \leq 5} \zeta^i v_i$, the residual $\zeta v_1 + B(q_5, q_5) - q_5$ has exactly the degrees 6–10 and local norm at most $\delta_5(\zeta)$, and $J_5 = 2B(q_5, \cdot)$ has norm at most $\ell_5(\zeta)$. For $0 < \zeta \leq \alpha_5$ a Neumann inverse and a binary-tree majorant give $v = q_5 + w$ with $\|w\|_{\text{loc}} \leq w_* = \frac{3}{4}(1 - \ell_5 - \sqrt{\mathcal{D}_5})$, the small root of $\frac{2}{3}w^2 - (1 - \ell_5)w + \delta_5 = 0$, endpoint included. The bounds are uniform in the box: each v_n is a sum of universal cluster functions over plaquette multisets, so the anchored sums of the infinite lattice bound every box; the orientation dependence of the Fourier-algebra norm is handled by taking the maximum over the eight axis reflections.

Proposition 3.5 (spectral return). *Under the analytic core, every eigenvalue $\lambda > 0$ of $H_L - E_0$ on the physical space satisfies $\lambda \geq \kappa d_5(\xi)$.*

Status. Proved here, given the analytic core; the workbench’s F35–F37, audited.

Proof. Let φ be a physical eigenvector, $h = \varphi/\psi$ (gauge invariant) and $h' = Q_H h \neq 0$ (as $\lambda \neq 0$, h is not constant). By Lemma 3.4 and $\Gamma(v, 1) = 0$, $(K - \lambda/\kappa)h' = 2Q_H \Gamma(v, h')$. Since h is smooth, $\|Kh'\|_X = \sum_j c_j \|A_j(h')\|_1$ is finite, by Plancherel and the Cauchy–Schwarz inequality on the finite product (the workbench’s F36); this is used below when dividing by it. If $\lambda/\kappa \geq 3$ there is nothing to prove, since $d_5 \leq 3$. Otherwise $c_j \geq 3$ on every nonconstant physical block (Lemma 3.2), so $c_j - \lambda/\kappa \geq c_j(1 - \lambda/(3\kappa))$ and $\|(K - \lambda/\kappa)h'\|_X \geq (1 - \lambda/(3\kappa))\|Kh'\|_X$. The core gives $\|2\Gamma(v, h')\|_X \leq 4t(v)\|Kh'\|_X \leq \chi\|Kh'\|_X$ with $\chi = 4 \sum t_i \xi^i + \frac{2}{3}w_*$, and Q_H does not increase $\|\cdot\|_X$. Hence $\lambda \geq 3\kappa(1 - \chi)$, and substituting w_* shows $3(1 - \chi) = d_5(\xi)$. $\square$

Nothing in this argument uses the truncation order.

Corollary 3.6 (the gap at every truncation order). *For $1 \leq N \leq 5$ let $\ell_N(x) = \sum_{i \leq N} (m_i + 4t_i)x^i$, $\delta_N(x) = \sum_{i,j \leq N, i+j \geq N+1} 3(m_i t_j + m_j t_i)x^{i+j}$, $\mathcal{D}_N = (1 - \ell_N)^2 - \frac{8}{3}\delta_N$, α_N the first positive root of $\mathcal{D}_N$ and $d_N(x) = \frac{3}{2}(1 + \sqrt{\mathcal{D}_N(x)}) + \sum_{i \leq N} (\frac{3}{2}m_i - 6t_i)x^i$. If (m_i, t_i) , $i \leq N$, bound the local norms of the first N vacuum coefficients, then $\Delta_L \geq \kappa d_N(\xi)$ for every $L \geq 2$, $a > 0$ and $0 < \xi \leq \alpha_N$, and d_N decreases on $[0, \alpha_N]$. In particular:*

(a) ($N = 1$) $\mathcal{D}_1(\xi) = 1 - 256\xi/3$ exactly and $\frac{3}{2}m_1 - 6t_1 = 0$, so

$$\Delta_L \geq \frac{3}{2}\kappa \left(1 + \sqrt{1 - \frac{64}{3g^4}}\right) \quad \text{for all } g^2 \geq 8/\sqrt{3} \approx 4.6188;$$

so $\Delta_L \geq \frac{3}{2}\kappa$ there, with 2.0745κ at $g^2 = 5$ and 2.8304κ at $g^2 = 10$.

(b) ($N = 2$) With the workbench’s pair $(m_2, t_2) = (\frac{5834}{39}, \frac{137}{6})$: $\alpha_2 = 0.0154800849822\dots$, that is $g^2 \geq 4.0186791538833\dots$, and $\Delta_L \geq \kappa d_2(\xi) \geq \kappa d_2(\alpha_2) = 1.52094\kappa$ on that range (1.7665κ at $g^2 = 4.1$, 2.3033κ at $g^2 = 5$). With the first pass’s recomputed values, rounded up to $(m_2, t_2) = (134.0674, 19.7954)$: $\alpha_2 = 0.0157977\dots$, that is $g^2 \geq 3.97807\dots$, with $\Delta_L \geq 1.52054\kappa$ on that range and $\Delta_L \geq 1.64707\kappa$ at $g^2 = 4$.

(c) ($N = 4$) $\alpha_4 = 0.0181049722316\dots$, that is $g^2 \geq 3.7159603625\dots$, with $d_4(1/64) = 1.898118$ and $d_4(4/225) = 1.658436$.

Status. Proved here from the audited core. (a) is the workbench’s own order-1 theorem, in its 14–16 September continuation YM-06 (consolidation/20260916, equations G22 and G26–G29, with $\Delta_L > 2.07445\kappa$ at $g^2 \geq 5$), whose own proof was not read; its input, order 1, is verified by hand. (b) is Generalised here relative to the workbench’s single-norm order-2 theorem in the same continuation (R10: $\Delta_L \geq \frac{3}{2}\kappa(1 + \sqrt{1 - \frac{256}{3}\xi + \frac{10720}{9}\xi^2})$ for $g^2 \geq \sqrt{(32 + \sqrt{354})/3} \approx 4.1156$). With the workbench’s two-norm pair and no further input, the threshold falls to 4.0187, and the endpoint constant rises from $\frac{3}{2}$ to 1.52094. The first-pass audit (46a_) computed $m(v_2) = 134.0673186784$ and $t(v_2) = 19.7953312398$ from explicit tensors. The referee pass of this reader found the closed forms $m(v_2) = 6 + \frac{11200}{117} + \frac{112}{9} + \frac{448\sqrt{3}}{39}$ and $t(v_2) = \frac{3}{2} + \frac{80}{9} + \frac{256}{39} + \frac{64\sqrt{3}}{39}$, which agree with them to ten digits. Rounded up (with $\sqrt{3} < 1.7320509$), they give the second threshold 3.9781 (recomputed here by root isolation). The same continuation reaches $g^2 \geq 3.8260$ at order 2 (its L17) by a sharper computed bound for the cubic term of the residual (944984/351 in its L10, against the generic $6(m_1 t_2 + m_2 t_1) = 300672/39$), which was not audited. So the qualitative content of Theorem 3.1 holds with audited inputs for $g^2 \geq 3.9781$, which includes the workbench’s benchmark $g^2 = 4$, and orders 3–5 lower the threshold to 3.6836 and raise the constants. (c) is conditional on the

order-3 and order-4 inputs and does not use order 5. α_N , the thresholds and the values of d_N computed here from the exact rationals of F26 by root isolation; the monotonicity checked on a 401-point grid for $N = 1, \dots, 5$. The workbench's YM-03 edition uses the radius $3/256 = \alpha_1$ for its row remainder.

Proof. Replace q_5 by $q_N = \sum_{i \leq N} \zeta^i v_i$ in the analytic core. By $v_n = \sum_{i+j=n} B(v_i, v_j)$ for $2 \leq n \leq N$, the residual is $\zeta v_1 + B(q_N, q_N) - q_N = \sum_{i,j \leq N, i+j \geq N+1} \zeta^{i+j} B(v_i, v_j)$, of local norm at most $\delta_N(\zeta)$, and $\|2B(q_N, \cdot)\| \leq \ell_N(\zeta)$. On $[0, \alpha_N]$, $\ell_N < 1$: $\ell_N(0) = 0$, and $\ell_N(\zeta_1) = 1$ with $0 < \zeta_1 \leq \alpha_N$ would give $\mathcal{D}_N(\zeta_1) = -\frac{8}{3}\delta_N(\zeta_1) < 0$. So the Neumann inverse and the tree majorant give $\|w\|_{\text{loc}} \leq w_* = \frac{3}{4}(1 - \ell_N - \sqrt{\mathcal{D}_N})$, and Proposition 3.5 with $\chi = 4 \sum_{i \leq N} t_i \zeta^i + \frac{2}{3}w_*$ gives $\lambda \geq 3\kappa(1 - \chi) = \kappa d_N(\zeta)$. Differentiating $\frac{2}{3}w_*^2 - (1 - \ell_N)w_* + \delta_N = 0$ gives $w'_* \sqrt{\mathcal{D}_N} = \delta'_N + \ell'_N w_* > 0$ on $(0, \alpha_N)$, so χ increases and d_N decreases. For $N = 1$, $(m_1, t_1) = (\frac{64}{3}, \frac{16}{3})$ gives $\ell_1 = \frac{128}{3}\zeta$, $\delta_1 = 6m_1 t_1 \zeta^2 = \frac{2048}{3}\zeta^2$ and $\mathcal{D}_1 = 1 - \frac{256}{3}\zeta + (\frac{16384}{9} - \frac{16384}{9})\zeta^2$; with $\zeta = 1/(4g^4)$, $\frac{256}{3}\zeta = \frac{64}{3g^4}$. For $N = 2$, $\frac{3}{2}m_2 - 6t_2 = \frac{3408}{39} > 0$, so $d_2 > \frac{3}{2}$ on the whole range even before the value $d_2(\alpha_2)$ is computed. $\square$

The same argument gives three further statements. They were found by the referee pass of this reader; the proofs below were checked here.

Corollary 3.7 (the gap on the full space). *Under the hypotheses of Corollary 3.6, every eigenvalue $\lambda > 0$ of $H_L - E_0$ on the whole space L^2 , not only on the physical space, satisfies $\lambda \geq \frac{1}{4}\kappa d_N(\zeta)$. This is sharp at $\zeta = 0$, where the gap on L^2 is $\frac{3}{4}\kappa$: one link in spin $\frac{1}{2}$.*

Proof. The ground state $e^v/\|e^v\|$ is also the ground state on L^2 : it is positive, and the positive ground state of H_L is unique. Lemma 3.4 holds for every h , gauge invariant or not. The bound $\|2\Gamma(v, h)\|_X \leq 4t(v)\|Kh\|_X$ of the core (the workbench's F10) uses only $\sum_e j_e \leq \frac{2}{3}c_j$ on the blocks of h , which holds on every block because $j(j+1) \geq \frac{3}{2}j$ for $j \geq \frac{1}{2}$. On L^2 the nonconstant blocks have $c_j \geq \frac{3}{4}$ instead of 3, so the proof of Proposition 3.5 gives $\lambda \geq \frac{3}{4}\kappa(1 - \chi) = \frac{1}{4}\kappa d_N(\zeta)$. $\square$

Status. Proved here. The record states the case $N = 1$ (web continuation, (G27), p. 66).

Proposition 3.8 (the gap at complex coupling). *Suppose the inputs of Corollary 3.6 are valid through order N , and let $\zeta \in \mathbb{C}$ with $|\zeta| \leq \alpha_N$. For every $L \geq 2$, on the physical space:*

- (a) $H_L(\zeta)e^{v(\zeta)} = E_0(\zeta)e^{v(\zeta)}$ with $E_0(\zeta) = \kappa(2M_L\zeta - P_H\Gamma(v, v))$, which is analytic in $|\zeta| < \alpha_N$;
- (b) every other eigenvalue $E_0(\zeta) + \lambda$ of $H_L(\zeta)$ has $\text{Re } \lambda \geq \kappa d_N(|\zeta|)$;
- (c) $E_0(\zeta)$ is an algebraically simple eigenvalue.

Hence the ground-state Riesz projection of $H_L(\zeta)$ is analytic in $|\zeta| < \alpha_N$, for every L . By contrast, the global route of Proposition 7.1 certifies only $|\zeta| < 3/(4M_L)$, which is at most $1/320$.

Proof. (a) The fifth-reference manuscript states the core for complex ζ with $|\zeta| \leq \alpha$ (its text after (F29), and (F31) with $x = |\zeta|$), and that the same exponential solves the complex equation (after (F34)). Lemma 3.4 holds for complex v . The coefficients v_n and the correction w are analytic in ζ , being locally uniform limits of polynomials. (b) Let φ be an eigenvector with eigenvalue $E_0 + \lambda$, $\lambda \neq 0$, and $h' = Q_H(e^{-v}\varphi) \neq 0$. As in Proposition 3.5, $(K - \lambda/\kappa)h' = 2Q_H\Gamma(v, h')$. For every block with $c_j \geq 3$ and $\text{Re } \lambda \geq 0$, $|c_j - \lambda/\kappa| \geq c_j - \text{Re } \lambda/\kappa \geq c_j(1 - \text{Re } \lambda/(3\kappa))$. If $\text{Re } \lambda < 0$, then $|c_j - \lambda/\kappa| \geq c_j$. With $\|2\Gamma(v, h')\|_X \leq \chi(|\zeta|)\|Kh'\|_X$ and $\chi < 1$, the case $\text{Re } \lambda < 0$ is impossible. Otherwise $\text{Re } \lambda \geq 3\kappa(1 - \chi) = \kappa d_N(|\zeta|)$. (c) The argument of (b) with $\lambda = 0$ gives $h' = 0$, so the eigenspace of E_0 is spanned by e^v . A generalised eigenvector $e^v h$ with $(H_L - E_0)(e^v h) = e^v$ would satisfy $\kappa(Kh - 2\Gamma(v, h)) = 1$. Applying Q_H gives $Kh' = 2Q_H\Gamma(v, h')$, so $\|Kh'\|_X \leq \chi\|Kh'\|_X$ and $h' = 0$. Then h is constant, $Kh = 0$ and $\Gamma(v, h) = 0$, and the equation reads $0 = 1$. The

family $H_L(\zeta) = \kappa K + \kappa\zeta(2M_L - S)$ is holomorphic of type (A) with compact resolvent, since S is bounded. The simple eigenvalue $E_0(\zeta)$ is separated from the rest of the spectrum by the strip $0 < \operatorname{Re}(z - E_0) < \kappa d_N(|\zeta|)$, so its Riesz projection is analytic (Kato [5], Ch. VII). $\square$

Status. Proved here, given the core; its complex form is the workbench’s own statement.

What the theorem is and is not. It is a finite-regulator, strong-coupling statement: fixed spacing, any box, bare coupling above a threshold. The continuum limit needs $a \rightarrow 0$ with $g \rightarrow 0$, the opposite regime, and the workbench records that its standing continuum path eventually leaves the domain. Nothing about the continuum mass gap follows, and the workbench claims nothing. Mass gaps at strong coupling are classical for lattice gauge theories in the Euclidean setting (Osterwalder–Seiler [9]; cited, not re-read); the workbench’s statement is an explicit bound, uniform in the box, on the whole physical space of the Kogut–Susskind $SU(2)$ Hamiltonian on open boxes, with a closed coupling threshold; whether a bound of this form is in the literature was not searched. If m_5 and t_5 were both underestimated by a factor 2, the threshold would move to $g^2 \approx 3.737$ and $d_5(1/64)$ to 1.8946; by Corollary 3.6(b), errors of any size in the order-3 to order-5 inputs leave a gap above 1.52κ for $g^2 \geq 3.9781$. The Riemann, Jacobian and S^6 material plays no role in this theorem.

3.3 The coupling domain and the continuum path

Proposition 3.9 (the standing continuum path leaves the domain). *Write $c = 1/g^2$. Then $\xi = c^2/4$, so $\xi \leq \alpha_5$ exactly when $c \leq 2\sqrt{\alpha_5} = 0.27147 \dots$, and $|\xi| < 1/55$ exactly when $c < 2/\sqrt{55} = 0.26968 \dots$. On the workbench’s standing path $a_n = a_0 2^{-n}$, $c_n = g_0^{-2} + \beta n \log 2$ with $\beta > 0$:*

- (a) *the first condition fails for every $n > (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2)$, and the second for every $n \geq (2/\sqrt{55} - g_0^{-2})/(\beta \log 2)$;*
- (b) *so at most $1 + \lfloor (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2) \rfloor$ points of the path lie in the domain $\xi \leq \alpha_5$. Whatever a_0 and g_0 , every point in the domain has spacing $a_n \geq a_0 e^{-2\sqrt{\alpha_5}/\beta}$.*

Status. Proved here; the workbench states (a) (§H7, and the fifth-reference README.md); (b) was found by the referee pass of this reader.

Proof. $\xi = 1/(4g^4) = c^2/4$, and $c \mapsto c^2/4$ is increasing on $c > 0$. Along the path c_n increases linearly without bound. For (b), a point in the domain has $n \leq (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2) \leq 2\sqrt{\alpha_5}/(\beta \log 2)$, so $a_n = a_0 2^{-n} \geq a_0 e^{-2\sqrt{\alpha_5}/\beta}$. $\square$

So Theorem 3.1 and the heat results of Section 4 say nothing about the regime of the continuum limit, as the workbench says. This limits what the results are about; it does not affect them at a fixed regulator.

4 Heat flow and the complementary physical space

The second manuscript of the latest edition, HEAT_AND_COMPLEMENT.md (H1–H34; Appendix B of the fifth-reference reader), uses the control of the vacuum from Section 3 for two further purposes: box-independent bounds on the heat correlations of plaquettes, and bounds on how much the plaquette family mixes with the rest of the physical space. This section states both, with proofs of the steps that carry the constants.

4.1 Objects

Let $A = H_L - E_0$ on the centred physical space and $\rho = \psi^2$. Every statement below holds for every $L \geq 2$ and $a > 0$.

- *Centred plaquette states and their heat correlations:*

$$r_p = (W_p - \langle W_p \rangle_\rho) \psi, \quad \widehat{C}_{pq}(\tau; \xi) = \langle r_p, e^{-\tau A/\kappa} r_q \rangle, \quad \tau = \kappa t.$$

- *Row norm:* $\|B\|_{\text{row}} = \max_p \sum_q |B_{pq}|$.
- *Taylor coefficients:* C_0, C_2, C_4 are the coefficients of ξ^0, ξ^2, ξ^4 in $\widehat{C}$; the workbench computed them in its 21 September heat edition.
- *The transformed generator:* in the coordinates $f \mapsto \psi f$, Lemma 3.4 turns A/κ into $K - D_v$ with $D_v h = 2\Gamma(v, h)$.
- *The Fourier-algebra norm* is $\|\cdot\|_X$ of Section 3, and $\chi = 4 \sum_i t_i \xi^i + \frac{2}{3} w_*$ is the number from the proof of Proposition 3.5, so $3(1 - \chi) = d_5(\xi)$ and $\|D_v h\|_X \leq \chi \|Kh\|_X$.

4.2 A heat remainder independent of the box

Theorem 4.1 (heat remainder; H13). *For every $L \geq 2$, every $\tau \geq 0$ and every $|\xi| < 1/55$,*

$$\|\widehat{C}(\tau; \xi) - C_0(\tau) - \xi^2 C_2(\tau) - \xi^4 C_4(\tau)\|_{\text{row}} \leq \frac{2829}{13} e^{-13\tau/8} \frac{(55|\xi|)^6}{1 - (55|\xi|)^2}.$$

The workbench states this with the prefactor $67896/169 = 401.75 \dots$ in place of $2829/13 = 217.61 \dots$; its bound is also valid.

Status. Conditional; audited in part.

- *Conditional* on the analytic core of Theorem 3.1, and therefore on the same order-3 to order-5 inputs.
- *Checked here.* Every constant is re-derived below. The evenness and the Cauchy step are proved.
- *Read, not checked in every step.* The semigroup construction (H5–H7) and the analytic continuation to complex ξ are standard. Proposition 3.8 gives the spectral side of that continuation.
- *The prefactor.* It is sharpened here by the mean estimate of step (i), which was found by the referee pass of this reader.
- *Not needed.* The inherited matrices C_0, C_2, C_4 are not needed for the truth of the bound, only for its use.
- *The earlier edition's bounds.*
 - Its bound on the circle $3/256$, with prefactor 512 and rate $3/2$, is superseded.
 - Its second bound U29b (heat reader of 21 September, p. 37), on the circle $45/4096$ with prefactor $6184/25$ and rate $15/8$, is sharper than the present theorem for small $|\xi|$ once $\tau > 12.6$ ($\tau > 10.15$ with the old prefactor). It is superseded by the family of radii below, which at $R = 45/4096$ has rate 2.2588 and prefactor 85.0.

The proof is in four steps; the constants are collected in Proposition 4.2.

(i) *The mean and the semigroup* (H1–H7). Put $T = Q_H D_v K^{-1}$ and $p = P_H D_v K^{-1}$ on the nonconstant physical coefficients. Since the norm $\|\cdot\|_X$ adds the scalar and the nonconstant parts, $|py| + \|Ty\|_X \leq \chi \|y\|_X$. So $S = (I - T)^{-1}$ exists with $\|S\| \leq 1/(1 - \chi)$, and the mean is $\mu(F) = P_H F + p S Q_H F$. The workbench bounds $|(\mu - P_H)F|$ by $\chi \|Q_H F\|_X / (1 - \chi)$. The additive form of the inequality gives more:

$$|(\mu - P_H)F| \leq \chi \|Q_H F\|_X, \quad \text{so} \quad |\mu F| \leq \|F\|_X \text{ for every } \chi < 1.$$

Indeed, let $y = Q_H F$ and $z = Sy$, so that $z = y + Tz$. Then

$$|pz| \leq \chi \|z\|_X - \|Tz\|_X \leq \chi (\|y\|_X + \|Tz\|_X) - \|Tz\|_X = \chi \|y\|_X - (1 - \chi) \|Tz\|_X \leq \chi \|y\|_X.$$

(Found by the referee pass of this reader; checked here, also on random examples in ℓ^1 .) For $L_0 = (I - T)K$ and $h > 0$, coefficientwise $\|(I + hK)f\|_X = \|f\|_X + h\|Kf\|_X$. Hence

$$\|(I + hL_0)f\|_X \geq \|f\|_X + h(1 - \chi)\|Kf\|_X \geq (1 + 3h(1 - \chi))\|f\|_X,$$

using $c_j \geq 3$ on nonconstant physical blocks (Lemma 3.2). With the range condition (H5), the Lumer–Phillips theorem [7] gives a semigroup $E(\tau) = e^{-\tau L_0}$ with $\|E(\tau)\| \leq e^{-3(1-\chi)\tau}$.

(ii) *A formula for the correlations.* Let $\mathcal{T}(\tau)$ be the transformed heat semigroup, so that $\widehat{C}_{pq}(\tau) = \mu[(\mathcal{T}(\tau)F_p)W_q]$ with $F_p = W_p - \mu W_p$; here $\mu[\mathcal{T}(\tau)F_p] = \mu[F_p] = 0$ removes the second mean. Let $h_\tau = K^{-1}SE(\tau)W_p$ and $G_z = \sum_q z_q W_q$ with $|z_q| \leq 1$. Since W_p has Haar mean 0, the Poisson equation (H3) gives $(K - D_v)h_\tau = E(\tau)W_p - \mu(E(\tau)W_p) = \mathcal{T}(\tau)F_p$. For any h and G ,

$$\int e^{2v}(Kh - 2\Gamma(v, h))G dU = \int e^{2v} \Gamma(h, G) dU,$$

by moving one derivative $X_{e,a}$ onto $e^{2v}G$. Dividing by $\int e^{2v}$ gives $\sum_q z_q \widehat{C}_{pq}(\tau) = \mu\Gamma(h_\tau, G_z)$, and $\|Kh_\tau\|_X \leq \|S\| \|E(\tau)\| \|W_p\|_X \leq \frac{8}{1-\chi} e^{-3(1-\chi)\tau}$.

Proposition 4.2 (the constants of the heat bound; H9–H12). *In this setting:*

- (a) $\|\Gamma(h, G_z)\|_X \leq 32\|Kh\|_X$;
- (b) $|P_H\Gamma(h, G_z)| \leq \frac{1}{8}\|Kh\|_X$;
- (c) $\|\widehat{C}(\tau)\|_{\text{row}} \leq \frac{1+255\chi}{1-\chi} e^{-3(1-\chi)\tau}$, and a fortiori the workbench's $\frac{1+255\chi}{(1-\chi)^2} e^{-3(1-\chi)\tau}$;
- (d) at $|\xi| = 1/55$ one has $1/55 < \alpha_5$ and $d_5(1/55) = 1.637\dots > 13/8$, so $\chi < 11/24$. The prefactor is at most $\frac{1+255 \cdot 11/24}{13/24} = \frac{2829}{13}$ (the workbench's $\frac{67896}{169}$ with its form of (c)), and the rate is at least $\frac{13}{8}$.

Status. Proved here. The workbench's (H9)–(H12) state (a), (b), the weaker form of (c) and (d). The earlier heat reader of 21 September already derives 32 (its U28), the rule $\frac{1}{8}$, and the weaker form of (c) (its U29a, pp. 36–37); this reader writes out the derivation. The sharper form of (c) uses the mean estimate of step (i). The inequalities of (d) are checked in exact rational arithmetic (scripts `ymr_heat_constants.py` and `ymr_referee_checks.py`).

Proof. (a) By (F10), $\|\Gamma(h, G)\|_X \leq 2t(G)\|Kh\|_X$. The function W_q has one Fourier block, with spin $\frac{1}{2}$ on the four links of q , and its coefficient matrix has trace norm $2^{4-1} = 8$ by (F12): four fundamental occurrences, two cyclic sign changes. A link lies on at most four plaquettes, so $t(G_z) \leq 4 \cdot \frac{1}{2} \cdot 8 = 16$. (b) Integration by parts gives $P_H\Gamma(h, G_z) = \int h KG_z dU = 3 \sum_q z_q \int h W_q dU$, since each W_q has Casimir $4 \cdot \frac{3}{4} = 3$. The gauge-invariant functions whose Fourier support is spin $\frac{1}{2}$ on the four links of q and spin 0 elsewhere form the line $\mathbb{C}W_q$: at each corner two spin- $\frac{1}{2}$ links meet, and $\frac{1}{2} \otimes \frac{1}{2}$ contains the trivial representation once. Write a_q for the component of h on this line. By Haar orthogonality and $\int W_q^2 = 1$, $\int h W_q = a_q$. The block of $a_q W_q$ contributes $3 \cdot 8|a_q|$ to $\|Kh\|_X$, so $|P_H\Gamma| \leq 3 \sum |a_q| \leq \frac{1}{8}\|Kh\|_X$. (c) $\mu\Gamma = P_H\Gamma + (\mu - P_H)Q_H\Gamma$, and $\|Q_H\Gamma\|_X = \|\Gamma\|_X - |P_H\Gamma|$. By step (i), then (b) and (a),

$$\begin{aligned} |\mu\Gamma(h_\tau, G_z)| &\leq (1-\chi)|P_H\Gamma| + \chi\|\Gamma\|_X \leq \left(\frac{1-\chi}{8} + 32\chi\right)\|Kh_\tau\|_X \\ &\leq \left(\frac{1-\chi}{8} + 32\chi\right) \frac{8e^{-3(1-\chi)\tau}}{1-\chi} = \frac{1+255\chi}{1-\chi} e^{-3(1-\chi)\tau}. \end{aligned}$$

With the workbench's form of step (i), the same steps give $\left(\frac{1}{8} + \frac{32\chi}{1-\chi}\right) \frac{8}{1-\chi} = \frac{1+255\chi}{(1-\chi)^2}$. The row norm is the maximum over the phases z_q of the left side. (d) The value $d_5(1/55)$ is computed from the rationals of (F26), and $d_5 > 13/8$ is equivalent to $\chi < 11/24$ since $d_5 = 3(1-\chi)$. Both prefactors increase with χ , and $3(1-\chi) > 3 \cdot \frac{13}{24} = \frac{13}{8}$. $\square$

(iii) *Evenness*. The map $U_{(n,i)} \mapsto (-1)^{\sum_{k<i} n_k} U_{(n,i)}$ multiplies each link by a central element. It preserves the Haar measure, K and the gauge action, and it sends every W_p to $-W_p$: for $p = (n; i, j)$ with $i < j$ the four signs multiply to -1 . It therefore conjugates $H_L(\xi)$ to $H_L(-\xi)$ plus a constant, and maps each r_p to $-r_p$ at the opposite coupling. So $\widehat{C}(\tau; -\xi) = \widehat{C}(\tau; \xi)$, and only even powers of ξ occur. (Checked on all 240 faces of the box $L = 2$.)

(iv) *Cauchy's estimate*. By (c) and (d), on the circle $|\xi| = 1/55$ every phase-weighted row sum of $\widehat{C}(\tau; \xi)$ is at most $Ce^{-13\tau/8}$ with $C = 2829/13$. So the coefficient of ξ^{2n} has row norm at most $Ce^{-13\tau/8}55^{2n}$, and summing over $n \geq 3$ gives the stated tail $\sum_{n \geq 3} (55|\xi|)^{2n} = (55|\xi|)^6 / (1 - (55|\xi|)^2)$. This proves Theorem 4.1.

The same step gives, for every n , $\|[\xi^{2n}] \widehat{C}(\tau)\|_{\text{row}} \leq \frac{2829}{13} 55^{2n} e^{-13\tau/8}$, and the analogous remainder after any even order. At $\xi = 0$ one has $\psi = 1$ and $r_p = W_p$, an eigenfunction of K with eigenvalue 3, so $C_0(\tau) = e^{-3\tau}I$ exactly. (Both noted by the referee pass.)

Other radii (Generalised here). Nothing in steps (i)–(iv) uses $R = 1/55$ except the numbers of Proposition 4.2(d). For every $0 < R \leq \alpha_5$ the same proof gives the bound of Theorem 4.1 with the following replacements: $55|\xi|$ by $|\xi|/R$; the prefactor by $C(R) = (1 + 255\chi_R)/(1 - \chi_R)$; the rate $13/8$ by $d_5(R) = 3(1 - \chi_R)$, where $\chi_R = 1 - d_5(R)/3$. At the endpoint $R = \alpha_5$, apply the Cauchy step at radii $r < R$ and let r increase to R , as the earlier edition does.

- At $R = 1/55$ the exact $\chi_R = 0.45412 \dots$ gives the prefactor $213.97 \dots$ and the rate $1.6376 \dots$; the rounding $\chi < 11/24$ costs little.
- At $R = \alpha_5$ the prefactor is $241.99 \dots$ and the rate $1.5453 \dots$
- At $R = 45/4096$, the radius of U29b, the prefactor is $85.00 \dots$ and the rate $2.2588 \dots$

With the workbench's form $(1 + 255\chi_R)/(1 - \chi_R)^2$ these prefactors are 391.98 , 469.79 and 112.89 . They are computed from the rationals of (F26); they are floating-point values, not rational certificates.

4.3 Relaxation into the whole complementary space

Let R be the map with columns r_p and $\Phi = \kappa A^{-1}R$. The original Grams and the energy of the family are

$$G_0 = R^*R, \quad G_1 = R^*\Phi, \quad G_2 = \Phi^*\Phi, \quad E = \kappa G_1, \quad K_0 = \kappa^{-1}R^*AR.$$

Put $P = \Phi G_2^{-1} \Phi^*$, $Q = I - P$, $B_\Phi = QR$ and $D_Q = QAQ$ on $Q\mathcal{H}_0$. Allowing the family Φc to relax into every $h \in Q\mathcal{H}_0$ gives the relaxed energy and Gram

$$E_{\text{full}} = E - \kappa^2 B_\Phi^* D_Q^{-1} B_\Phi, \quad G_{\text{full}} = G_2 + \kappa^2 B_\Phi^* D_Q^{-2} B_\Phi.$$

Proposition 4.3 (relaxation bounds; H20–H29). *Suppose that, for numbers $w_0, w_1, w_2, w_K, \beta \geq 0$ and $d_{\text{ph}} > 0$,*

$$\|G_k - 3^{-k}I\| \leq w_k \quad (k = 0, 1, 2), \quad \|K_0 - 3I\| \leq w_K, \quad 0 \leq J = G_0 - 6G_1 + 9G_2 \leq \beta I, \quad A \geq \kappa d_{\text{ph}}.$$

Put $l_k = 3^{-k} - w_k > 0$, $u_k = 3^{-k} + w_k$ and $u_K = 3 + w_K$. Then:

- the fraction of each Φc seen by the plaquette states, $\|P_R \Phi c\|^2 / \|\Phi c\|^2$, is at least $\max\{l_1^2 / (u_0 u_2), 1 - \beta / (9l_2)\}$, where P_R is the orthogonal projection onto the span of the r_p ;
- $(1 - \eta_E)E \leq E_{\text{full}} \leq E$ and $G_2 \leq G_{\text{full}} \leq (1 + \eta_G)G_2$, with $\eta_E = \beta / (d_{\text{ph}} l_1)$ and $\eta_G = \beta / (d_{\text{ph}}^2 l_2)$;
- the energy of $(I - P_R)\Phi c$ is at most $(u_1 u_K / l_0^2 - 1)$ times $c^* E c$.

Status. Proved here; the workbench's (H21), (H23), (H26)–(H29), with the second bound of (a) added. The domain statement for D_Q (H25) is the workbench's and was read.

Proof. (a) $\|P_R \Phi c\|^2 = c^* G_1 G_0^{-1} G_1 c \geq u_0^{-1} c^* G_1^2 c \geq (l_1^2/u_0)|c|^2$ and $\|\Phi c\|^2 = c^* G_2 c \leq u_2 |c|^2$. For the second bound (found by the referee pass): $G_1 = \kappa R^* A^{-1} R$ is symmetric, so $J = (R - 3\Phi)^*(R - 3\Phi)$, and $\|(I - P_R)\Phi c\|^2 \leq \|\Phi c - \frac{1}{3}Rc\|^2 = \frac{1}{9}c^* J c \leq \frac{1}{9}\beta |c|^2$, while $\|\Phi c\|^2 \geq l_2 |c|^2$. (b) $B_\Phi^* B_\Phi = G_0 - G_1 G_2^{-1} G_1$ is the least value of $(R - \Phi X)^*(R - \Phi X)$ over coefficient matrices X , and $X = 3I$ gives J ; so $B_\Phi^* B_\Phi \leq J \leq \beta I$. Since $D_Q \geq \kappa d_{\text{ph}}$, $\kappa^2 B_\Phi^* D_Q^{-1} B_\Phi \leq (\kappa/d_{\text{ph}})\beta I \leq (\beta/(d_{\text{ph}} l_1))\kappa G_1$, and $\kappa^2 B_\Phi^* D_Q^{-2} B_\Phi \leq (\beta/d_{\text{ph}}^2)I \leq (\beta/(d_{\text{ph}}^2 l_2))G_2$. (c) Since $A\Phi = \kappa R$ and $P_R R = R$, the mixed energy is $q_A(\Phi c, P_R \Phi c) = \kappa \langle Rc, \Phi c \rangle = \kappa c^* G_1 c = q_A(\Phi c)$, and $P_R \Phi c = R G_0^{-1} G_1 c$ has energy $\kappa c^* G_1 G_0^{-1} K_o G_0^{-1} G_1 c$. Expanding $q_A((I - P_R)\Phi c)$ gives $\kappa c^* (G_1 G_0^{-1} K_o G_0^{-1} G_1 - G_1) c$, the workbench's (H29). Now $G_1 G_0^{-1} K_o G_0^{-1} G_1 \leq (u_K/l_0^2)G_1^2 \leq (u_K u_1/l_0^2)G_1$. $\square$

The workbench obtains the numbers w_k, w_K and β from Theorem 4.1, and w_K from a bound on the kinetic rows, which carries the constants 30 and 48 (H14).

Proposition 4.4 (the kinetic rows; H14). *For each plaquette p and each complex $|\zeta| \leq 1/55$, the entries $\mu\Gamma(W_p, W_q)$ of the kinetic matrix satisfy:*

- (a) $\Gamma(W_p, W_p) = 3 - \chi_1(U_p)$, of norm $\|\cdot\|_X = 3 + 27 = 30$;
- (b) for each of the 12 plaquettes q adjacent to p , $\Gamma(W_p, W_q) = \frac{3}{4}P_0 - \frac{1}{4}P_1$. Here P_0, P_1 are the parts of $W_p W_q$ with spin 0 and spin 1 on the shared link, of Casimirs $\frac{9}{2}$ and $\frac{13}{2}$. So $\|\Gamma(W_p, W_q)\|_X$ is 24 for 8 of the q and $6 + 6\sqrt{3}$ for the other 4; both are below 48. For non-adjacent $q \neq p$, $\Gamma(W_p, W_q) = 0$;
- (c) hence $\sum_q |\mu\Gamma(W_p, W_q)| \leq 3 + \chi(27 + 192 + 24 + 24\sqrt{3}) < 134$ for $\chi < \frac{11}{24}$. The workbench's bound is $30 + 12 \cdot 48 = 606$.

Proof. From $2\Gamma(f, h) = (c_f + c_h)fh - K(fh)$ for block eigenfunctions (the workbench's F17):

- *Diagonal.* $W_p^2 = 1 + \chi_1(U_p)$ and $\chi_1(U_p)$ has Casimir $4 \cdot 2 = 8$, so $2\Gamma(W_p, W_p) = 6(1 + \chi_1) - 8\chi_1$. The norm of $\chi_1(U_p)$ is $3^3 = 27$ by F12 with $d = 3$.
- *Adjacent faces.* $2\Gamma(W_p, W_q) = 6(P_0 + P_1) - \frac{9}{2}P_0 - \frac{13}{2}P_1$.
- *The block norms.* The trace norms of P_0 and P_1 were computed from explicit $SU(2)$ tensors in the first-pass audit (46a₋). They are (16, 48) for two thirds of the adjacent pairs and $(8, 24\sqrt{3})$ for the rest, which gives $\frac{3}{4} \cdot 16 + \frac{1}{4} \cdot 48 = 24$ and $\frac{3}{4} \cdot 8 + \frac{1}{4} \cdot 24\sqrt{3} = 6 + 6\sqrt{3}$. The referee pass of this reader found the split $8 + 4$ for the twelve neighbours of one face.
- *Non-adjacent faces.* For non-adjacent $q \neq p$ no link is shared, so no derivative acts on both factors, and $\Gamma = 0$.
- *The row.* Only the diagonal entry has a constant part, namely 3. So step (i) gives $|\mu\Gamma| \leq |P_H\Gamma| + \chi\|Q_H\Gamma\|_X$ entrywise, and the sum follows. $\square$

Status. Proved here, given the first pass's block norms; the sharper row sum found by the referee pass. The workbench's constants 30 and 48 are thereby derived, and 48 is not attained.

The time-integral formula $G_k = \int_0^\infty \tau^{k-1} \widehat{C}(\tau) d\tau / (k-1)!$ ($k \geq 1$) gives the remainder factors C/d^k with $d = 13/8$. For $J = \widehat{C}(0) + \int_0^\infty (9\tau - 6)\widehat{C}(\tau) d\tau$ the signed weight is kept: $\int_0^\infty |9\tau - 6|e^{-d\tau} d\tau = \frac{6}{d} - \frac{9}{d^2} + \frac{18}{d^2}e^{-2d/3}$, and with $e^{-13/12} < \frac{339}{1000}$ the prefactor is $C_J = 5156090136/3570125$ (H17–H18). The degree-2 and degree-4 parts are the workbench's inherited row bounds (the table after H18), which come from 199 supports and 559 marked contributions of its earlier volume edition.

Proposition 4.5 (the benchmark table; H5). *From the stated inherited row bounds and the constants C, C_J and 606, the widths and Proposition 4.3 give, for every $L \geq 2$ and $a > 0$:*

Coupling	Observed fraction	Energy loss η_E	Gram increase η_G
$g^2 \geq 10$	> 0.9778	< 0.01057	< 0.01123
$g^2 \geq 12$	> 0.9975	< 0.00115	< 0.00120
$g^2 \geq 25/2$	> 0.9984	< 0.00071	< 0.00073
$g^2 \geq 13$	> 0.99904	< 0.000436	< 0.000451
$g^2 \geq 16$	> 0.99991	< 0.0000357	< 0.0000364

In particular the workbench's table holds: at $g^2 \geq 13$ the energy loss and the Gram increase are below $1/2000$, and at $g^2 \geq 16$ below $1/25000$.

Status. Reproduced: recomputed in exact rational arithmetic from the stated inputs (script `ymr_heat_constants.py`); at $g^2 = 13$, with the workbench's $d_{\text{ph}} = 29047/10000$, the three values agree with its diagnostics to ten digits. At the other couplings the script uses d_{ph} equal to $d_5(\xi)$ rounded down to four digits. Conditional on the inherited degree-2 and degree-4 row bounds and on the inputs of Theorem 3.1. The widths are monotone in ξ and d_5 decreases in ξ , so each row holds for all larger g^2 , as the workbench says.

With the sharper constants of this reader the same computation gives more. These are the prefactor $2829/13$ of Theorem 4.1, the leakage prefactor $C_J = 782.29 \dots$ (the workbench's C_J times $13/24$), the kinetic bound 134 of Proposition 4.4, and the second bound of Proposition 4.3(a):

Coupling	Observed fraction	Energy loss η_E	Gram increase η_G
$g^2 \geq 10$	> 0.99458	< 0.005713	< 0.006053
$g^2 \geq 12$	> 0.99940	< 0.000623	< 0.000647
$g^2 \geq 25/2$	> 0.99963	< 0.000380	< 0.000394
$g^2 \geq 13$	> 0.99977	< 0.000237	< 0.000244
$g^2 \geq 16$	> 0.999981	< 0.0000194	< 0.0000198

(Script `ymr_referee_checks.py`; the referee pass computed the same values independently.)

The plaquette states also bound the gap from above. This was found by the referee pass; the proof was checked here.

Proposition 4.6 (an upper bound on the gap). (a) For every $L \geq 2$, $a, g > 0$ and every plaquette p ,

$$\Delta_L \leq \kappa \frac{3 - \langle \chi_1(U_p) \rangle_\rho}{1 + \langle \chi_1(U_p) \rangle_\rho - \langle W_p \rangle_\rho^2}.$$

(b) With the widths of Proposition 4.3, $\Delta_L/\kappa \leq (3 + w_K)/(1 - w_0)$. With the constants above, the gap therefore lies in

- $[2.8382, 3.0055]\kappa$ for $g^2 \geq 10$;
- $[2.9047, 3.00024]\kappa$ for $g^2 \geq 13$;
- $[2.9372, 3.00003]\kappa$ for $g^2 \geq 16$.

The lower ends are $d_5(\xi)$, under the inputs of Theorem 3.1.

(c) At each fixed L , the first-order term of Δ_L at $\xi = 0$ vanishes. By (b), $\Delta_L \leq 3\kappa + O(\xi^2)$ uniformly in L . Every lower bound d_N of Corollary 3.6 is $3 - 64\xi + O(\xi^2)$, so its first-order loss is an artefact of the method.

Proof. (a) The state r_p is gauge invariant and orthogonal to ψ . By the min–max principle $\Delta_L \leq q_A(r_p)/\|r_p\|^2$. By (F3), $q_A(r_p) = \kappa \langle \Gamma(W_p, W_p) \rangle_\rho = \kappa(3 - \langle \chi_1 \rangle_\rho)$ (Proposition 4.4(a)). And $\|r_p\|^2 = \text{Var}_\rho(W_p) = 1 + \langle \chi_1 \rangle_\rho - \langle W_p \rangle_\rho^2$, since $W_p^2 = 1 + \chi_1(U_p)$. (b) The two quantities in (a) are

$(K_o)_{pp} \leq 3 + w_K$ and $(G_o)_{pp} \geq 1 - w_0$. (c) At $\xi = 0$ the eigenvalue 3κ of A on the physical space has the eigenspace spanned by the W_p : by Lemma 3.2, $c_j = 3$ only for spin $\frac{1}{2}$ on a cycle of length 4, that is, on a plaquette. The first-order perturbation matrix is $\langle W_q, SW_{q'} \rangle = \sum_r \int W_q W_r W_{q'} dU = 0$. Indeed, two distinct faces share at most one link, so in each such product some link occurs exactly once, in spin $\frac{1}{2}$; and $\int W_q^3 = 0$. The ground energy has no first-order term either, since $\int S = 0$. Finally, w_K and w_0 are $O(\xi^2)$, and $d'_N(0) = -\frac{3}{2}\ell'_N(0) + \frac{3}{2}m_1 - 6t_1 = -64$. $\square$

Status. Proved here; (a) is unconditional, (b) rests on the inputs of Proposition 4.5. The windows are computed by `ymr_referee_checks.py`.

4.4 Heat horizons, support quotients and the spatial limit

- *Finite heat time* (H33; the Gram comparison is the workbench's, the choice of T is checked here). Replacing Φ by $\Phi_T = (I - e^{-TA})\Phi$ changes the Grams by factors between $(1 - \varepsilon)^2$ and 1, with $\varepsilon = e^{-\kappa d_{\text{ph}} T}$. For a signed sum of log det terms of total rank B the error is at most $2B(-\log(1 - \varepsilon)) \leq 2B\varepsilon/(1 - \varepsilon)$, and the choice $T = (\kappa d_{\text{ph}})^{-1} \log(1 + 2B/\eta)$ makes $\varepsilon/(1 - \varepsilon) = \eta/(2B)$, so the error is at most η .
- *Support quotients* (H31–H32; located). An exact isomorphism between quotients of inverse-energy spans and kernels, and the attained minimum of the family energy over changes of support.
- *The spatial limit at fixed spacing* (H34; located, an outline). Heat coefficients through degree N agree on two boxes containing the same N -neighbourhood of the marks, so Theorem 4.1 makes the correlations converge as $L \rightarrow \infty$; the workbench then builds the infinite-volume semigroup, its invariant state and the relaxation bounds. This is a limit at fixed a and g , not a continuum limit, as the workbench says.

5 The earlier editions

5.1 The continuations of 14–21 September

Their gap and heat bounds are superseded by Sections 3 and 4; their other calculations remain in the record as dated editions.

- *14–16 September* (the 297-page web-continuation reader). The first gap theorems of the same architecture:
 - a gauge-native order-1 theorem (equations G22 and G26–G29), with $\Delta_L > 2.07445\kappa$ at $g^2 \geq 5$;
 - a single-norm order-2 theorem (R10), for $g^2 \geq \sqrt{(32 + \sqrt{354})/3} \approx 4.1156$;
 - an order-2 theorem (L17) for $g^2 \geq 3.8260$, which uses a computed bound $944984/351$ for the cubic term of the residual.

Status. Located; the proofs were not read.

- The order-1 theorem is Corollary 3.6(a), proved there.
- R10 is strengthened by Corollary 3.6(b): the threshold is 4.0187 with the workbench's order-2 pair, and 3.9781 with the first pass's order-2 values.
- L17's computed bound was not checked.
- One arithmetic line of this edition is wrong. It justifies the self bound $40/27$ of its table (C21) by " $38/81 + 1564/810 = 40/27$ " (p. 153), but the left side equals $12/5$. The value $40/27 = 3 \cdot \frac{1}{81} \cdot 8 + 15 \cdot \frac{1}{810} \cdot 64$ is the one that the referee pass of this reader obtained for the self cluster, so the bound itself stands (found by the referee pass; the arithmetic checked here).

- *17 September* (the 49-page quartic cube reader). The fourth-order vacuum source in two presentations, equal as polynomials (78 representatives, 8,621 anchored multisets); the sixth-order vacuum energy, with the contribution $-83/1944$ of the elementary cube; and the gap bound $\Delta_L > 1.6584\kappa$ for $g^2 \geq 15/4$.

Status. The gap bound is Corollary 3.6(c) at $\xi = 4/225$, conditional on the order-3 and order-4 inputs; Theorem 3.1 gives $d_5(4/225) > 1.6993$ there. The rest located. The workbench records that this edition keeps its earlier review qualification and that the 21 September heat proof does not recertify the earlier interval (yang-mills/README.md).

- *18–19 September* (inside the cumulative archives). The sixth vacuum source, the eighth-order energy, and a finite response estimate on the 240 faces of the box $L = 2$.

Status. Located. The workbench says that the new spatial tables and the final execution package were not supplied, so their completeness is reported by the source, not replayed.

- *21 September, heat and volume* (the 46-page reader). The fourth-order heat coefficients (84 time functions over 17 geometric cases); a box-independent heat remainder on the circle $|\xi| < 3/256$; the fixed-spacing spatial limit; and the full-complement energy retained above 999/1000 at $g^2 \geq 16$.

Status. Superseded quantitatively by Section 4: radius $1/55 > 3/256$, and a loss below $1/25000$ against $1/1000$. This edition also has a second remainder bound, U29b (p. 37), on the circle $45/4096$ with prefactor $6184/25$ and rate $15/8$. It is superseded by the family of radii of Section 4, not by Theorem 4.1 alone. The five proof manuscripts are located. One of them receives the zeta programme’s Gram-sandwich lemma; its steps T1–T3 were checked in 21_ (Section 8).

5.2 The foundations of 8–9 September

The three manuscript lines of §2.2 contain the workbench’s first results. They are stated for the full finite Hamiltonian, with its scalar constant, open boundaries and all gauge constraints, and they say what they do not prove (§2.3).

- *Quantum line* (79 pages).
 - Exact extensive blocking maps, and volume-independent lower bounds on the high-energy spectral weight of selected dressed Wilson-loop channels, at every positive coupling (§11).
 - A first volume-uniform gap, $\Delta \geq \kappa(3 - 512\xi)$ for $0 < \xi \leq 1/49152$, uniform in the box and the block size (§10, equation (77)). It is proved there with the creation method of Yarotsky [12], with Baez’s spin-network framework [2] for the physical space (p. 19). The spatial line uses it for its fixed-coupling results (spatial §28: $g_0^4 \geq 12288$, which is the same condition).
 - A nonzero non-Abelian vertex on the box $L = 2$ (§17).
 - A seven-dimensional lowest-band map that is a bijection for small coupling g , at a fixed box.
 - The Jacobian-map and Navier–Stokes branches (Section 6).
- *Spatial line* (129 pages). Sequences in which the box, the spacing and the coupling vary together; its conclusion (§28) states what they prove.
 - Along a fixed-coupling diagonal with $g_0^4 \geq 12288$, the magnetic excitation probability eventually has zero weight on every bounded energy interval.
 - For fixed Wilson-loop states, a positive high-energy mass and a failure of strong continuity.
 - On a selected weak-coupling diagonal, the finite gaps close while the corrected magnetic probability escapes to high energy.
 - The collapse of the finite-box analyticity disk (§27; Proposition 7.1).

- *Volume line* (81 pages). Volume-independent local estimates for the actual vacuum. They include:
 - local electric moments at every coupling;
 - the exact free physical threshold 3κ (§15);
 - estimates on the S^6 -derived cusp at $0 < \xi \leq 10^{-16}$;
 - §26: along the logarithmic coupling path, the upper end of its certified spectral window diverges. The document calls this a limitation of its own estimate, not a statement about the spectrum.

Status. Located, apart from the items proved or audited elsewhere in this reader: §10's gap (Proposition 5.1), §27 (Proposition 7.1), and Theorems 13.1 and 14.2 of the quantum line (Section 6).

Proposition 5.1 (the first volume-uniform gap is implied). *For $0 < \xi \leq 3/256$ the order-1 bound of Corollary 3.6(a) satisfies $d_1(\xi) \geq 3 - 128\xi$. So the quantum line's $\Delta \geq \kappa(3 - 512\xi)$ on $0 < \xi \leq 1/49152$ follows from Corollary 3.6(a), which holds on the larger range $\xi \leq 3/256$ ($g^2 \geq 8/\sqrt{3}$).*

Status. Proved here.

Proof. With $y = 256\xi/3 \in (0, 1]$, $d_1(\xi) = \frac{3}{2}(1 + \sqrt{1-y}) \geq \frac{3}{2}(2-y) = 3 - 128\xi$, since $\sqrt{s} \geq s$ for $0 \leq s \leq 1$. And $1/49152 < 3/256$. $\square$

6 The side branches: the Jacobian map, Navier–Stokes and S^6

Three branches of the record use material from outside lattice gauge theory. None of them enters the results of Sections 3 and 4. The attribution file of 21 September says so for the 21 September heat and volume proofs. For the fifth-reference results it follows from their proofs above, which use only the Hamiltonian, its Fourier calculus and the computed coefficients. This section states what each branch proves in this record and how far it was checked. The statements and statuses come from the reader of four workbenches [11], where they were refereed, and from the notes cited.

6.1 The Jacobian map and the low-mode theorem

The map

$$F(x, y, w) = ((1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), 2x - 3x^2 y - x^3 w)$$

has $\det DF = -2$ and three preimages of $(-\frac{1}{4}, 0, 0)$, namely $(0, 0, -\frac{1}{4})$ and $(\pm 1, \mp \frac{3}{2}, \frac{13}{2})$ (checked symbolically). The workbench's attribution file identifies it with the counterexample to the Jacobian conjecture announced by Levent Alpöge on 20 July 2026, whose announcement, as the file records, credits Akhil for the question and Fable for the work; it notes that the S^6 construction is a different object. That Keller maps give incompressible polynomial flows, and that the curve $\gamma(\tau) = (z^{-1}, -3z/2, 13z^2/2)$, $z = \sqrt{1 - 8\tau}$, escapes at $\tau = 1/8$, is proved in the zeta reader [13] (Lemma 6.6, “Keller maps give incompressible polynomial flows”). The workbench's chain $F \rightarrow$ incompressible flow $\rightarrow$ material tensor $\rightarrow$ lattice local-energy weights $\rightarrow$ trial states ends in its Theorem 14.2, audited in 34_: it is a valid diagonal argument, conditional on fixed-box weak-coupling limits that the file only sketches, and its energy quotients equal, up to a factor $1 \pm 1/(10j)$, the lowest free two-gluon colour-singlet energy of an open box of side $j/50$ at couplings $g_j < 1/j$. It bears on the mass gap in neither direction, and the data of F move the quotients only inside a window that does not depend on them.

6.2 Navier–Stokes

In this record the Navier–Stokes material enters through the quantum line’s §§20–22: the velocity field of a solution becomes a reducible $SU(2)$ connection $A_i = \lambda u_i T$, with curvature density $-2 \sum_{i < j} \text{tr} F_{ij}^2 = \lambda^2 |\text{curl } u|^2$, and gauge-invariant curvature states built from it are followed along a weak-coupling diagonal.

- *The imported theorem* (not audited). The source is OpenAI’s manuscript *Finite Time Blowup for Navier–Stokes* [8], Theorem 1.1: for every viscosity, a compactly supported smooth force and zero initial velocity give a smooth solution on $[0, 1)$ with bounded energy and $\limsup \|u(t)\|_\infty = \infty$. The record’s first README.md says that source statements and targeted derivations were inspected, and that no full independent analytic or Lean verification of the external existence proof is claimed. Every use of it is conditional on it.
- *The map to Yang–Mills* (audited, conditional; 21_ §2). The identity above and the constants of the curvature masses were checked. The zero-quotient diagonal exists only along couplings $g \rightarrow 0$, and it rests on imported rate bounds for the Navier–Stokes solution and a dissipation bound, which were not checked. The source calls it “a rigorous weak-coupling counterexample candidate, not a completed Millennium disproof”.

The workbench’s separate Navier–Stokes collection (the repository’s `navier-stokes/`, with its own Zenodo record 10.5281/zenodo.22678406) is not part of this record; its results are stated in the reader of four workbenches [11].

6.3 S^6

The S^6 construction is the manuscript [1]: a compact complex threefold fibred by tori over the projective line, which the manuscript states is diffeomorphic to S^6 . That statement was not checked here, and the workbench does not certify a global complex structure on S^6 ; it draws no mass-gap conclusion. Its attribution file, which it designates as the clarification, says that the manuscript was circulated by Levent Alpöge and produced with Claude (older files say “with Fable”).

In this record the construction enters only through its regular-fibre period matrix. The spatial and volume lines use its imaginary-period block to build a magnetic background, magnetic link transports and a gauge-covariant map into the Wilson Hilbert space (the attribution file’s “period-dependent magnetic branch”; volume line §1 and §19). On the patch that these lines use, only the combination $D = L_T q_T + 6m_T^2$ of the imaginary periods enters, with $L_T = \text{Im } \tau_T$, $q_T = -\text{Im } \beta_T$ and $m_T = \text{Im } \mu_T$ (28_, Lemma 28.2), through the angle $\theta = 2\pi a^2/D$. So the Yang–Mills calculations need one positive number from the S^6 data, and nothing about S^6 itself; the attribution file says the same about its hypotheses.

The S^6 project’s own results, and the record in which they are published (10.5281/zenodo.22678442), are the subject of a separate reader.

7 Negative results, with exact scope

The record contains several obstructions: proved limits of particular methods, and sequences shown not to give what was sought. Each is stated here with what it does not show. None bears on whether the continuum theory has a mass gap.

7.1 The global perturbative route

Proposition 7.1 (the collapse of the global analyticity disk; spatial line §27). *Write $H = (2g^2/a)(H_0 - \xi W_L) + \text{const}$ on the full space (no gauge projection), with $H_0 = \sum_e E_e$ the sum of the link Casimirs E_e and $W_L = \sum_p W_p$. The operator-norm Neumann series certifies the isolated rank-one ground-state projection on the circle $|z| = 3/8$ exactly for $|\xi| < 3/(16M_L) = 1/(64L^2(2L + 1))$, and on*

no contour separating 0 from the rest of the spectrum does the norm-product criterion give more. On the physical space the same route gives exactly $|\zeta| < 3/(4M_L)$, on the circle $|z| = 3/2$.

Status. Proved here; the full-space statement audited (YM-07 §27); the physical-space radius and the optimality over contours are Generalised here.

Proof. $|W_p| \leq 2$ with equality at $U \equiv I$, so $\|\zeta W_L\| = 2|\zeta|M_L$; the same holds on the physical space, since for every $\varepsilon > 0$ the indicator of the nonempty open gauge-invariant set $\{W_L > 2M_L - \varepsilon\}$ is a physical vector. The spectrum of H_0 is contained in $\{0\} \cup [3/4, \infty)$, so $\|(H_0 - z)^{-1}\| \leq 8/3$ on $|z| = 3/8$, and the series converges when $2|\zeta|M_L \cdot \frac{8}{3} < 1$. Both factors are attained ($\|(H_0 - z)^{-1}\| = 8/3$ at $z = \pm 3/8$), so the norm-product criterion fails beyond this radius. Any contour separating 0 from the rest of the spectrum crosses $(0, \frac{3}{4})$, since the eigenvalue $\frac{3}{4}$ must have winding number 0; at a crossing point x the resolvent norm is $\max\{1/x, 1/(\frac{3}{4} - x)\} \geq \frac{8}{3}$. On the physical space the spectrum of H_0 lies in $\{0\} \cup [3, \infty)$ by Lemma 3.2, with 3 attained by one plaquette in spin $\frac{1}{2}$, and the same two arguments give the resolvent norm $\frac{2}{3}$ on $|z| = \frac{3}{2}$ and the radius $3/(4M_L)$. $\square$

The radius shrinks with the box, so at any fixed coupling this route eventually fails; it does not show that the true analyticity radius is small, and it proves neither a gap nor its absence. It explains why Theorem 3.1 needs local norms: at $L = 2$ ($M_2 = 240$) the global route certifies only $|\zeta| < 1/1280$ on the full space and $|\zeta| < 1/320$ on the physical space, both far inside $\zeta \leq \alpha_5 \approx 1/54.3$. (“Weak-coupling disk” in the source means small $|\zeta|$, that is large g : the regime of Theorem 3.1.)

7.2 The other obstructions

Source	What fails	Status	What it does not show
fifth reference (§H7), and Proposition 3.9	the strong-coupling estimates along the standing continuum path	proved here	anything about the continuum regime, in either direction
fifth reference (§H7)	re-centring the elementary (order-1, single-norm) certificate to extend the source domain: it returns its old endpoint $3/256 = \alpha_1$	located (the workbench’s ROUTE_ASSESSMENT calculation)	that other routes fail; the workbench names the next candidate
quantum line §11	lower bounds for dressed channels as bounds for the whole spectral measure or for all physical states	the document’s own statement	anything about the gap
volume line §26	the certified spectral window of the weighted covariance stays bounded along the logarithmic path: its upper end diverges	located	that the covariance has no weight in a fixed window
spatial line §§17–18	a selected weak-coupling diagonal as the sought interacting continuum: the finite gaps close while the magnetic probability escapes	located	the prescribed logarithmic and fixed-electric-coefficient paths, or other continuations
spatial line §20	a unique-vacuum realisation of a selected macroscopic limit of a second observable ($d\delta_0$, $d = 1 - 2^{-3/2}$)	located	anything beyond that selected limit

Source	What fails	Status	What it does not show
quantum line, Theorem 14.2 (Jacobian branch)	a mass-gap conclusion from the trial states: their energies are the free two-gluon threshold of a shrinking-coupling box	audited in 34_ (Section 6)	anything about the mass gap
quantum line §§20–22 (Navier–Stokes branch)	a zero energy quotient at fixed coupling: the zero-quotient sequence exists only along couplings tending to 0	checked in 21_ §2, conditional on the imported rate bounds	whether another state with vanishing quotient exists

All of these concern methods, sequences or observables. The workbench states each of them itself, in the passages cited; in the passages read, no negative result of the record is stated more broadly than its proof.

8 Where the workbench meets the author’s other workbenches

The note 47_ records every crossing between the owner’s workbenches that the audit found, as typed edges: *P*, a proof-level import; *D*, a dependency or citation; *I*, the same identity or object on both sides; *A*, an analogy only; *R*, a regression fixture; *M*, shared methodology. The edges that touch this record are these.

Edge	What crosses	Type	Status
$S^6 \rightarrow \text{YM}$	The imaginary period block of the construction’s period matrix, as a magnetic background; only $D = L_T q_T + 6m_T^2$ enters	P	audited (28_, Lemma 28.2)
$F \rightarrow \text{NS} \rightarrow \text{YM}$	The Jacobian map’s incompressible flow, its material tensor, and Theorem 14.2	P	audited (28_, 34_); conditional on fixed-box limits
$\text{zeta} \rightarrow \text{YM}$	The Gram sandwich $(1 - \delta)^2 \Phi^* \Phi \leq \Phi_J^* \Phi_J \leq (1 + \delta)^2 \Phi^* \Phi$ when $\ (\Phi_J - \Phi)x\ \leq \delta \ \Phi x\ $, $\delta \leq 1$; the YM instance takes $\delta = e^{-T\lambda_0}$ from the heat semigroup, where $\lambda_0 \leq A$ is a spectral lower bound ($\lambda_0 = 27\kappa/10$ in the 21 September edition; 21_ writes a_0)	P	audited (21_ §3; the condition $\delta \leq 1$ is necessary for the lower bound)
$\text{zeta} \rightarrow \text{YM}$	Inverse-power observability, and the composed minimum-lift identity	P (pat- terns)	arithmetic and algebra checked (46a_)
$\text{NS} \rightarrow \text{YM}$	The velocity field as a reducible $SU(2)$ connection, $-2 \sum \text{tr} F_{ij}^2 = \lambda^2 \text{curl } u ^2$	P; D on the im- ported rate bounds	audited, conditional (21_ §2)
Collatz $\rightarrow$ YM	The identity $X_v - X_u = (q - 3)(q + 4)/32$, used as a regression fixture only	R	verified (46a_)
ES $\leftrightarrow$ YM	The word “gap”, and a proposed success projector	A	no mathematical content

Edge	What crosses	Type	Status
YM $\leftrightarrow$ zeta	The workbench’s own summary of the zeta routes, with one transcription error ($2.098 \dots \cdot 10^{-11}$ written as 10^{-19} , in <code>ATTEMPTS.md</code> and <code>research-attempts.json</code>)	A	located

None of these crossings enters Theorem 3.1, Theorem 4.1 or Proposition 4.3: their proofs use only the Hamiltonian, its Fourier calculus and the computed coefficients. The zeta programme’s Gram lemma is used in the 21 September heat edition (Section 5), not in the fifth reference, whose own manuscript says that no arithmetic input is imported (H8).

9 What remains open, and what was not checked

9.1 Open questions

1. *The continuum.* Every result of the record holds at a fixed lattice spacing and at strong bare coupling (Proposition 3.9). A continuum theory, and its mass gap, are outside what the record addresses; the spatial line’s conclusion names the interacting local continuum and a nonzero low-energy spectral weight as its unresolved target (§28).
2. *Lower thresholds.*
 - *What the workbench shows.* Re-centring its elementary certificate, of order 1 and in a single norm, returns that certificate’s endpoint $3/256 = \alpha_1$ (`ROUTE_ASSESSMENT.md`; §H7).
 - *The next candidate.* The workbench names the same-support linearized action J_5 with a preconditioned residual. It records that the sixth-source catalogue is missing from the delivered sources (§H7).
 - *Open.* Can a re-centred certificate of order N reach past α_N if it uses data beyond the positive majorants, such as signed or same-support information? For certificates built only from majorant series with nonnegative coefficients, Pringsheim’s theorem suggests not; the workbench’s calculation shows this for $N = 1$.
3. *Independent confirmation of the inputs.* The threshold $g^2 \geq 3.6836$ rests on computed coefficient bounds of orders 3 to 5. An implementation independent of the workbench’s programs would remove the condition; with the inputs checked here the threshold is 3.9781 (Corollary 3.6(b)).
 - *Order 3, computed by the referee.* The referee pass of this reader computed the order-3 norms numerically, with its own implementation (Clebsch–Gordan contraction over the 612 anchored labels, in 22 symmetry classes). It found $m(v_3) = 1203.1424 \dots$ and $t(v_3) = 127.2886 \dots$, below the workbench’s 1611.59 and 180.35. Its checks include the record’s energy coefficient $e_4 = 5M/216 - 2J/1053$, reproduced exactly.
 - *What it would give.* If confirmed, preferably in exact arithmetic, the inputs of orders 1–3 give $\Delta_L \geq \kappa d_3(\xi)$ for $g^2 \geq 3.7154$ (the threshold is 3.71533...), with $d_3(1/64) = 1.902$.
 - *Status.* The computation of the norms is reported by the referee and was not repeated here. The threshold and d_3 were recomputed here from its values, rounded up to $(m_3, t_3) = (1203.1425, 127.2887)$.
 - *Order 4.* An independent computation of order 4 (8,621 anchored multisets) would need a spin-network implementation.

4. *The first-order loss.* Is there $c > 0$ with $\Delta_L \geq \kappa(3 - c\tilde{\zeta}^2)$ for all L and all small $\tilde{\zeta} > 0$? The true gap has no first-order term (Proposition 4.6(c)). The lower bounds lose one because the spectral return bounds $2\Gamma(v, h')$ by $4t(v)\|Kh'\|_X$.
5. *Clustering.* Suppose, as §H34 asserts, that the coefficient of $\tilde{\zeta}^n$ in $\widehat{C}_{pq}$ vanishes unless the face distance of p and q is at most n . Then the coefficient bounds of Section 4 give decay $|\widehat{C}_{pq}(\tau; \tilde{\zeta})| \leq \frac{2829}{13} e^{-13\tau/8} (55|\tilde{\zeta}|)^{2[d(p,q)/2]} / (1 - (55|\tilde{\zeta}|)^2)$, uniform in L . A written proof of that locality statement is open.

9.2 What this reader did not check

- *The computed inputs.* The coefficient logic of orders 3 to 5 (F14–F25) was not re-implemented. The first-pass audit re-ran the workbench’s producer and auditor, and their output was byte-identical to the published rows (Appendix B).
- *The inherited heat data.* The matrices C_0, C_2, C_4 and the degree-2 and degree-4 row bounds were not re-derived (Proposition 4.5). The kinetic-row constants are now derived (Proposition 4.4).
- *The spatial limit.* The domain argument (H25) was read. The spatial limit (H34) is an outline and is located.
- *The earlier editions.* The 8–9 September lines were read only in the passages cited, and the 14–19 September continuations were not read beyond their statements (Section 5).
- *The Overleaf archive.* The 87 MB archive 08_ was not downloaded.
- *The side branches.* The Navier–Stokes theorem imported from [8], and the global statement of [1], were not checked; neither is used by the results of Sections 3–4.
- *The literature cited above* (Osterwalder–Seiler, Schütte–Zheng–Hamer, Eymard, Lumer–Phillips, Yarotsky, Baez, Kato) was not re-read. No literature search for novelty was made. The results marked as found by the referee pass are short consequences of the record’s material and of standard tools, and no novelty is claimed for them.

A Catalogue of statements

Every statement of the record that this reader states, proves, reproduces or locates, with where it is in the record and how far it was checked, in the labels defined in “About this reader”. “F” and “H” refer to the two manuscripts of the fifth-reference edition; the audit’s inventory labels (YM-01 and so on, from 47a_) are given where earlier notes use them.

Statement	Where in the record	Status here	Here
<i>The strong-coupling gap (fifth reference)</i> $\Delta_L \geq \kappa d_5(\tilde{\zeta})$ for all $L \geq 2, a > 0$, $g^2 \geq 3.6836$ (YM-01)	F26–F42	Conditional on the order-3 to order-5 inputs; the argument audited	Thm 3.1
Gauge-invariance Casimir inequality	F7 (girth 4)	Generalised here, to every girth; the exact constant for each link (found by the referee pass)	Lemma 3.2, Rem. 3.3
Ground-state transform	F33–F36	Proved here, for complex v too	Lemma 3.4
Spectral return	F35–F37	Proved here, given the core	Prop. 3.5

Statement	Where in the record	Status here	Here
The gap at truncation orders 1, 2, 4; with audited order-2 values, $g^2 \geq 3.9781$	14–17 Sept. editions; F	Proved here from the audited core	Cor. 3.6
The gap on the full space, $\frac{1}{4}\kappa d_N(\xi)$	web continuation (G27) for $N = 1$	Proved here (found by the referee pass)	Cor. 3.7
The gap at complex coupling; analyticity of the ground state on $ \xi < \alpha_N$, for every L	F29–F34 (the complex core)	Proved here, given the core (found by the referee pass)	Prop. 3.8
The standing continuum path leaves the domains; at most $1 + \lfloor (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2) \rfloor$ points lie in them	§H7	Proved here (the count found by the referee pass)	Prop. 3.9
<i>Heat flow and the complementary space (fifth reference)</i>			
Box-independent heat remainder on $ \xi < 1/55$ (YM-02)	H1–H13	Conditional on the core; audited in part; prefactor sharpened to $\frac{2829}{13}$	Thm 4.1
Its constants $32, \frac{1}{8}, \frac{1+255\chi}{(1-\chi)^2}, \frac{67896}{169}, \frac{13}{8}$	H9–H12; earlier U28–U29a	Proved here; sharpened to $\frac{1+255\chi}{1-\chi}$ (found by the referee pass)	Prop. 4.2
The kinetic-row constants 30 and 48	H14	Proved here; sharpened row bound 134	Prop. 4.4
Evenness of the connected correlations	H12–H13	Proved here	§4
Relaxation bounds and observed fraction	H20–H29	Proved here, with a second bound for the fraction	Prop. 4.3
Benchmark table ($1/2000$ at $g^2 \geq 13$; $1/25000$ at $g^2 \geq 16$)	H5	Reproduced; conditional on the inherited row bounds; sharpened	Prop. 4.5
An upper bound on the gap; no first-order term	—	Proved here ((a) unconditional; found by the referee pass)	Prop. 4.6
Finite heat time	H33	Audited in part (the choice of time)	§4
Support quotients; the spatial limit at fixed spacing	H31–H32; H34	Located (H34 an outline)	§4
<i>Earlier editions</i>			
Order-1 and order-2 gap theorems (G22, G26–G29; R10; L17)	14–16 Sept. reader	Order 1 proved here; R10 strengthened; L17 located	§5
Fourth-order source, sixth energy ($-83/1944$), gap for $g^2 \geq 15/4$ (YM-05)	17 Sept. reader	The gap proved here, conditional (Cor. 3.6(c)); rest located	§5
Sixth source, eighth energy, 240-face response (YM-04)	18–19 Sept., in the archives	Located	§5
Heat on $ \xi < 3/256$ and on $ \xi < 45/4096$ (U29b); spatial limit; complement at $1/1000$ (YM-03)	21 Sept. heat reader	Located (superseded)	§5
First volume-uniform gap $\kappa(3 - 512\xi)$, $\xi \leq 1/49152$ (Yarotsky’s method)	quantum line §10, (77)	Proved here from Cor. 3.6(a)	Prop. 5.1

Statement	Where in the record	Status here	Here
Channel bounds, blocking maps, non-Abelian vertex, band bijection (YM-07)	quantum line	Located	§5
Spectral escape, loop states, selected diagonals, pure-gauge limit	spatial line §§1–26, 28	Located	§5
Volume-independent local estimates; free threshold 3κ ; cusp estimates	volume line	Located	§5
<i>Side branches</i>			
$\det DF = -2$ and the three-point fibre over $(-\frac{1}{4}, 0, 0)$	quantum line; attribution file	Reproduced (symbolically)	§6
Material-tensor sectors (Theorem 13.1)	quantum line	Audited in part (endpoint coefficients recomputed, 28_); the rest located	§6
Trial states at the free two-gluon threshold (Theorem 14.2)	quantum line	Audited (34_), conditional on fixed-box limits	§6
Navier–Stokes blowup theorem (imported)	quantum line §§20–22	Located (not audited)	§6
Velocity field as an $SU(2)$ connection; zero-quotient diagonal	quantum line §§20–22	Audited, conditional (21_)	§6
S^6 period data enter only through D	spatial and volume lines	Audited (28_)	§6
<i>Negative results</i>			
The global analyticity disk shrinks with the box (full and physical space, every contour)	spatial line §27	Proved here (full space audited; the rest generalised here)	Prop. 7.1
Re-centring the order-1 certificate returns its endpoint 3/256	§H7; ROUTE_ASSESSMENT.md	Located	§7
Channel bounds are not bounds on the whole measure	quantum line §11	The document’s own statement	§7
Certified window diverges on the logarithmic path	volume line §26	Located	§7
Gaps close on a selected diagonal; no unique vacuum for a selected limit	spatial line §§17–18, 20	Located	§7

B How this reader was checked

Sources.

- The repository KokunoYumeto/yang-mills-interacting-workbench at commit fa79faf, read-only.
- The record 10.5281/zenodo.22883643. Its file list and MD5 checksums, and the files added by each of its five versions, were read from the record’s public metadata on 26 September 2026. Twenty-three of its 43 files are byte-identical to files of the repository at fa79faf. Nineteen of the other twenty were downloaded from the record, with the owner’s permission, and their MD5 checksums equal the record’s. The 87 MB Overleaf archive was not downloaded. The script ymr_record_inventory.py reproduces this inventory.

What was refereed earlier. Section 3 and Proposition 7.1 come from the Yang–Mills section of the reader of four workbenches [11], in which they went through five stages:

1. a first-pass audit by a separate Claude instance (46a_). It recomputed the closed-form arithmetic of the theorem from the ten rationals of (F26), recomputed the order-2 inputs from explicit $SU(2)$ coefficient tensors, and checked order 1 by hand. It also re-ran the workbench’s order-5 producer (662 rows, 51 seconds) and its polynomial and dual-certificate auditor (20 seconds): both outputs were byte-identical to the published ones. The workbench’s execution record reports these runs; its publication intake matched their hashes but did not repeat them;
2. a referee pass on the notes 43_–47_, by an instance that had written none of them;
3. a referee pass on that reader;
4. two further passes that read it for understatement and missed generality as well as for overstatement. They produced the girth form of Lemma 3.2, Corollary 3.6 and the physical-space part of Proposition 7.1, and corrected the credit for the order-1 and order-2 theorems of the 14–16 September edition.

The checks are in `checks/workbenches/`: the first pass in `audit46_ym/`, then `workbench_reader_claims_checks.py` and `generality_checks.py`.

What this reader adds to the reader of four workbenches. “New” here means new relative to that earlier reader, not new mathematics.

- the map of the record and its five versions (Section 2);
- Section 4: Theorem 4.1 with its proof, and Propositions 4.2, 4.3 and 4.5;
- Propositions 3.9 and 5.1;
- the Navier–Stokes and S^6 parts of Section 6, and the table of Section 7.

These went through the referee pass described next.

The referee pass of this reader. An independent Claude instance (claude-opus-5-5), which wrote none of the text, refereed version 1 of this reader on 27 September 2026. A first attempt was cut off by a usage limit before it wrote a report; its programs are kept. The second attempt read the whole reader, both proof manuscripts in full and every record passage cited, in both directions, as the owner asked. It reran the reader’s two scripts, whose outputs were byte-identical, and wrote its own checks. It found no error that invalidates a stated result. It reported two major and fourteen minor findings, and results A–K. Every finding was verified here before it was applied: by reading the record at the cited place, or by an independent script (`ymr_referee_checks.py`, `ym_order2_threshold.py`).

- *Understatements.*
 - With the first pass’s order-2 values the audited threshold is 3.9781, not 4.0187 (Corollary 3.6(b)). The rational rounding had to be made with $\sqrt{3} < 1.7320509$; the referee’s $\sqrt{3} < 1.7321$ was not tight enough for the stated bounds.
 - The heat prefactor improves from 67896/169 to 2829/13 (Proposition 4.2).
 - The fraction bound gets a second form (Proposition 4.3(a)).
 - The kinetic-row constants are derived and sharpened (Proposition 4.4).
 - The benchmark table is sharpened accordingly.
- *Attributions and descriptions.*
 - The factor 255, the constant 32 and the rule $\frac{1}{8}$ are in the earlier heat reader.
 - That reader’s second bound U29b is described.

- The order-5 replay was reported by the workbench, and its intake did not repeat it.
 - The attribution file’s statement concerns the 21 September heat and volume proofs.
 - Yarotsky and Baez are credited.
 - Five of the twenty Zenodo-only files are inside committed archives.
 - The re-centring result is the order-1 certificate’s.
 - The status labels are made consistent, and the scripts are described as they are.
- *Results adopted.* After verification: Remark 3.3, Corollary 3.7, Propositions 3.8, 4.4 and 4.6, the count in Proposition 3.9(b), the remarks after the Cauchy step, and the record’s arithmetic erratum in C21.
 - *Reported, not repeated.* The referee’s numerical computation of the order-3 norms is reported in Section 9 and was not repeated here. Its one-loop reading of the path coefficient β was not adopted.

The referee’s report and its programs are in `checks/ym_reader/referee/`.

The scripts of this reader. They import no workbench code. They are in `checks/ym_reader/` on the branch named below.

Script	What it checks	Time, result
<code>ymr_heat_constants.py</code>	the arithmetic of the manuscript’s constants (H9)–(H13), including $d_5(1/55) > 13/8$ in exact arithmetic from the rationals of (F26); the centre map on all 240 faces of $L = 2$; the signed-weight integral (H17) and C_J (H18); the widths and the workbench’s benchmark table (H19)–(H30) at five couplings, with its diagnostics at $g^2 = 13$ reproduced to ten digits; the endpoints of Proposition 3.9; the constants on other radii (30 checks). In version 2 a vacuous check was replaced	1 s; all pass
<code>ymr_referee_checks.py</code>	the findings adopted from the referee pass: the order-2 threshold 3.9781; the mean lemma on random examples; $2829/13$, C_J , the other radii and U29b; the kinetic row; the sharpened table and the gap windows; the identity for J ; the erratum in C21; $d'_N(0) = -64$; the order-3 threshold arithmetic (32 checks)	1 min; all pass
<code>ymr_record_inventory.py</code>	the 43 files of the record against the repository and the downloaded copies, by MD5	under 1 s; 23 + 19, one not local

What has not been done. This version has had one independent referee pass. The results adopted from it were proved by the referee and checked here, but they have not had a second independent reading. Nothing in this reader has been reviewed by a human specialist. The parts of the record marked located were not checked, and §9 lists what else was left out.

Where everything is. This reader, its sources and its scripts are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/`: the reader in `ym_reader/`, the scripts in `checks/ym_reader/`, and the audit notes at the top level.

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